

Long–Short Portfolio Return Book

Cumulative return of each factor's long–short portfolio, net of 10 bp transaction cost.
272 factors over 134 months, ordered by in-sample L–S CAGR (last 48 months, net of cost).

HOW TO READ

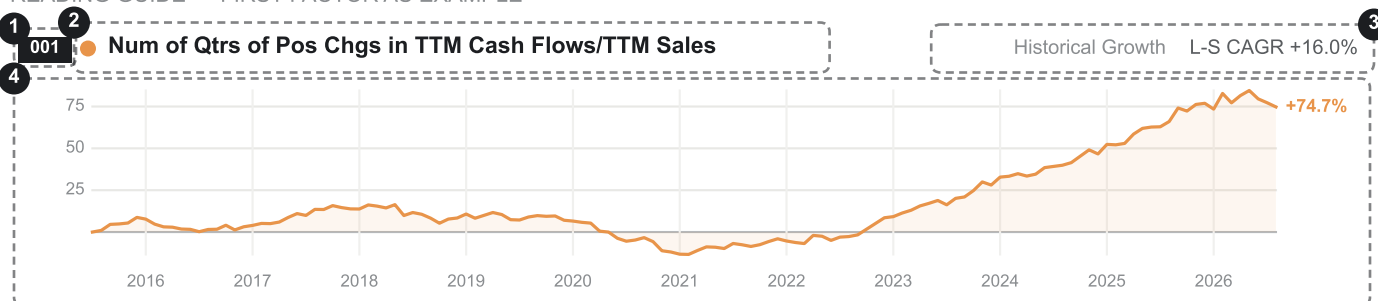
Each curve starts at 0 % in July 2015 and compounds monthly long–short returns. The line is coloured by the factor's style; the figure at the right edge is the period-end cumulative return.

FACTOR STYLES

Valuation	42	Price Momentum	54
Earnings Quality	55	Size	3
Analyst Expectations	24	Historical Growth	56
Capital Efficiency	28	Volatility	10

001 In current model portfolio (50 factors)

READING GUIDE — FIRST FACTOR AS EXAMPLE



1 Rank - ordered by ③ L-S CAGR, descending.
Filled dark badge = in the current Top-50 portfolio.

2 Factor name - colour dot = style (legend above).

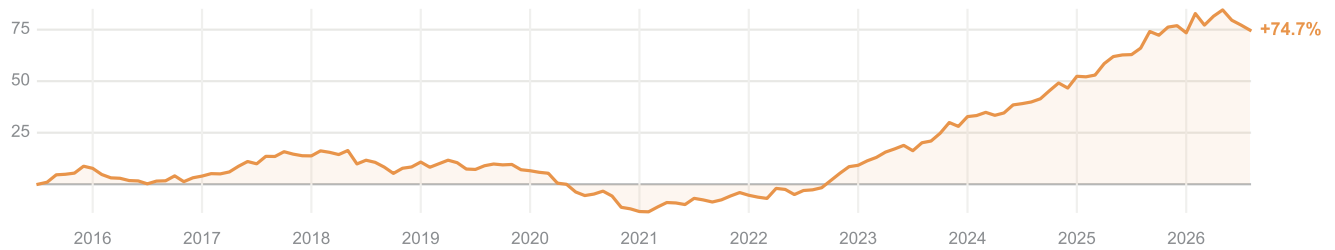
3 Style name / L-S CAGR - the LAST 48 MONTHS only (rolling IS), net of 10 bp cost. The live selection / sort basis. The curve below uses these same live 48M L/S rules.

4 Curve - the live 48M L/S rules applied over FULL HISTORY (Jul 2015 - Aug 2026), net of 10 bp cost. Rules are fit on recent data, so the long-run curve is context, not a live track record - it can rise while the recent ③ is negative.

001 ● Num of Qtrs of Pos Chgs in TTM Cash Flows/TTM Sales

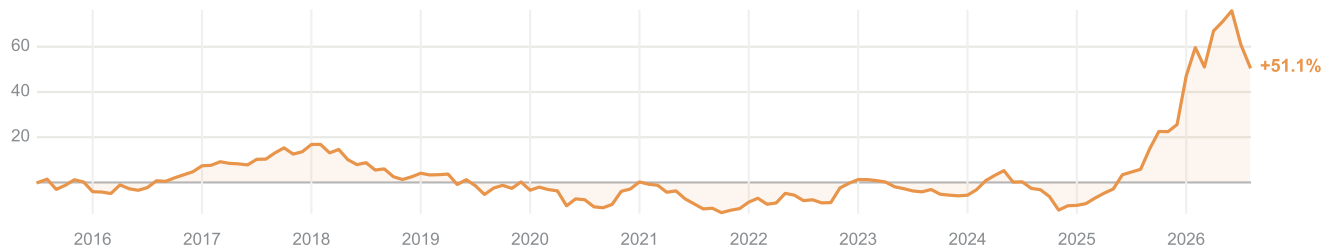
Historical Growth

CAGR +16.0%

**002 ● Trailing 12-Month Receivables and Inventories/Trailing 12-Month Sales**

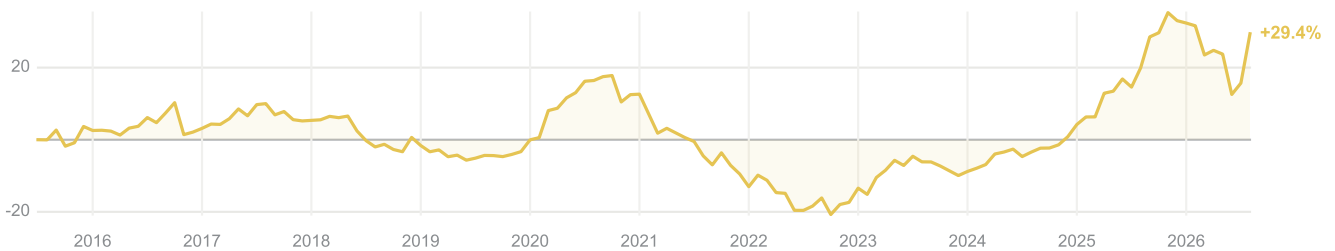
Historical Growth

CAGR +13.7%

**003 ● Long Term Debt to Equity Ratio**

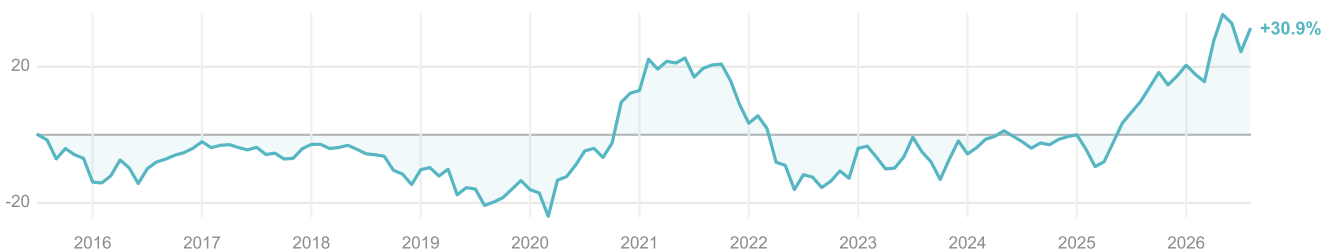
Capital Efficiency

CAGR +12.1%

**004 ● 12M Realized Price Volatility**

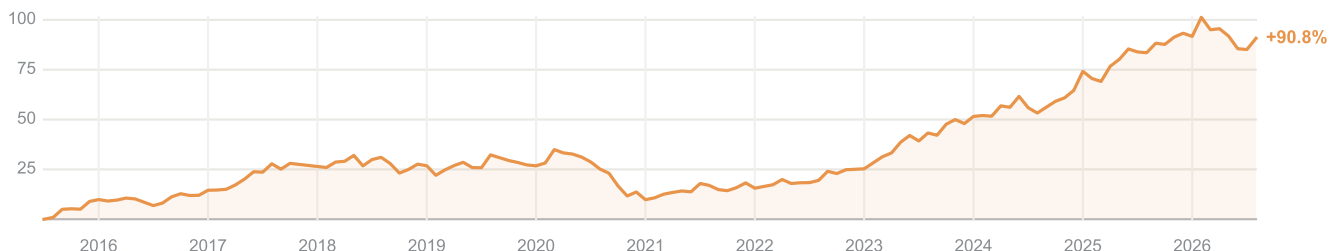
Volatility

CAGR +11.9%

**005 ● Number of Consecutive Qtrs of Pos. Changes in TTM Cash Flows**

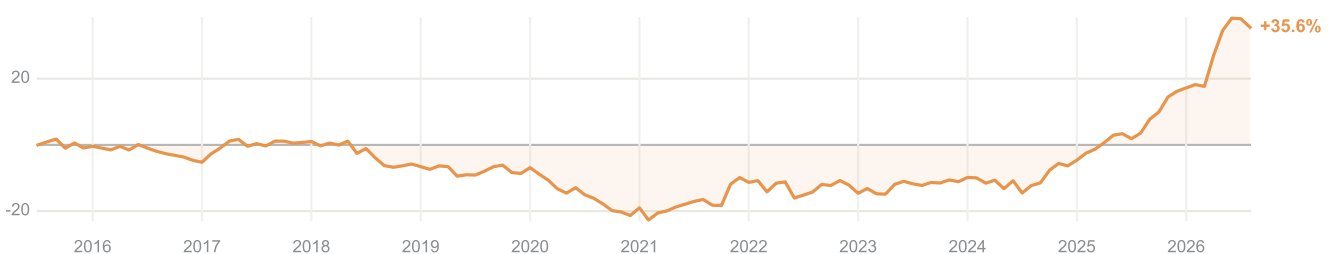
Historical Growth

CAGR +11.7%

**006 ● Change of Asset Turnover**

Historical Growth

CAGR +11.2%



007 ● **Analyst Dispersion for FY2 EPS**

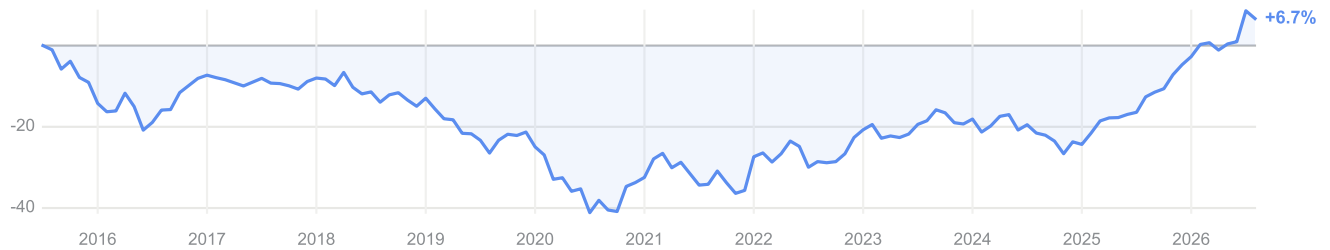
Analyst Expectations

CAGR +11.2%

008 ● **Assets to Price Ratio**

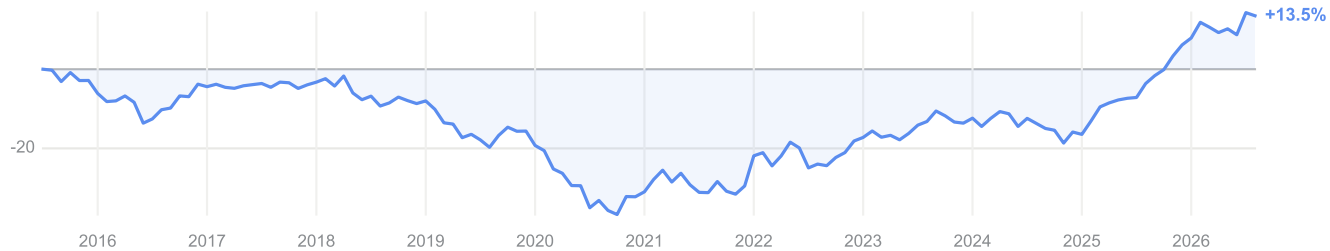
Valuation

CAGR +10.9%

009 ● **Current Liability to Price**

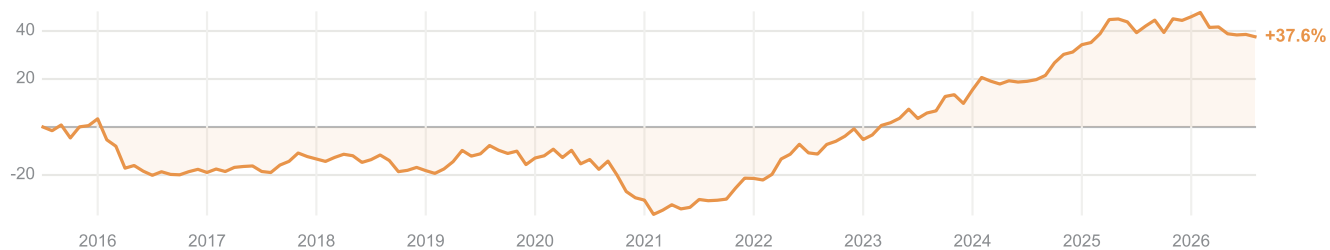
Valuation

CAGR +10.9%

010 ● **Number of Consecutive Qtrs of Positive Changes in TTM Quarterly Earnings**

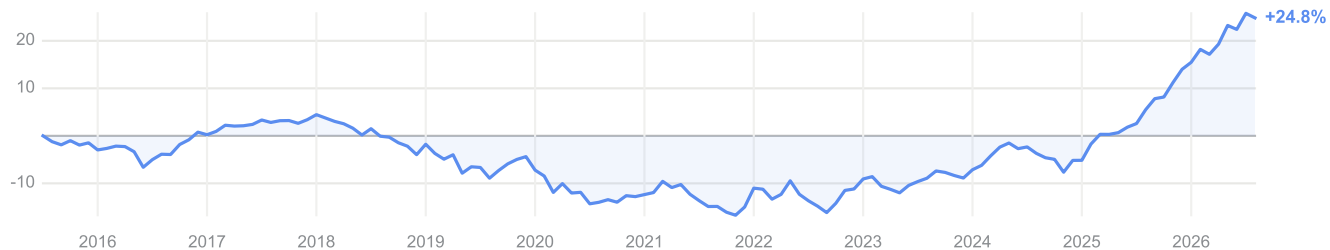
Historical Growth

CAGR +10.6%

011 ● **Inverse P/E Ratio Adj for Growth and Yield**

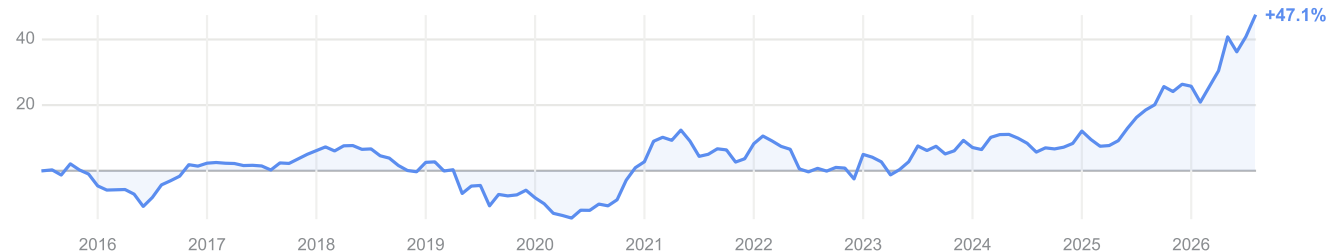
Valuation

CAGR +10.5%

012 ● **Cash to Adjusted Enterprise Value**

Valuation

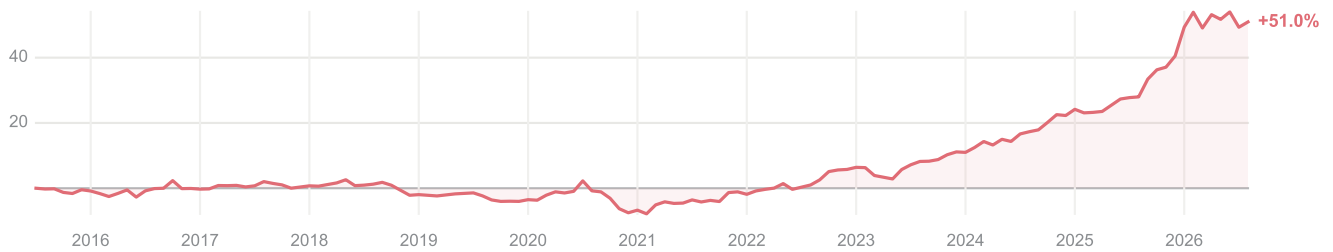
CAGR +10.4%



013 ● Unexpected Inventory Change

Earnings Quality

CAGR +10.4%

**014 ● Operating Earnings to Assets Ratio**

Earnings Quality

CAGR +10.4%

**015 ● Adjusted 3M Revision in FY1 EPS Estimate**

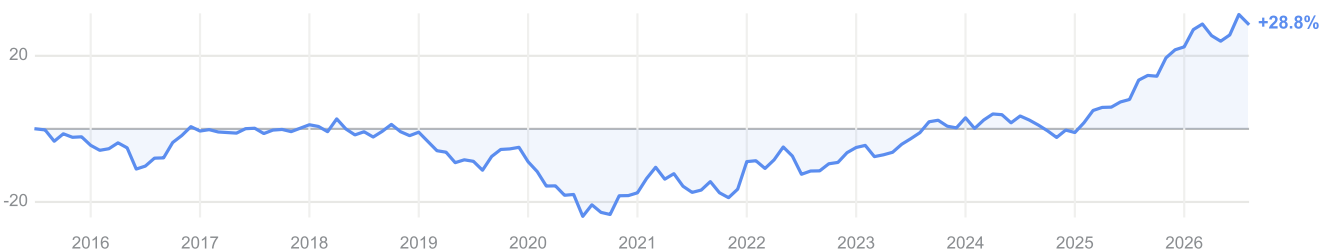
Analyst Expectations

CAGR +10.1%

**016 ● Sales to Price Ratio**

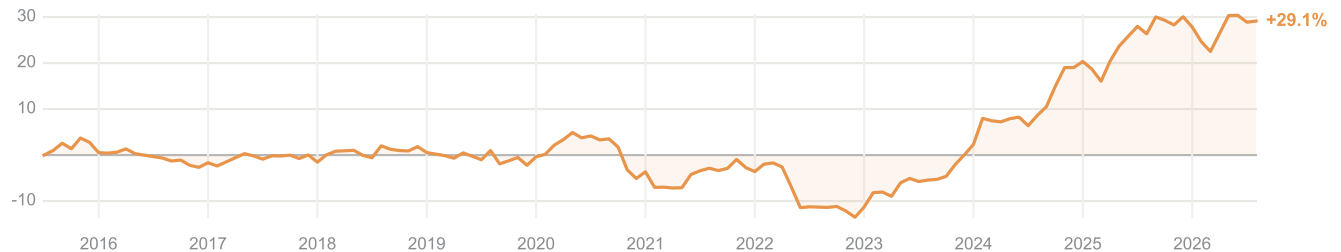
Valuation

CAGR +10.1%

**017 ● Adj 1Y Chg in Asset Adjusted FCF**

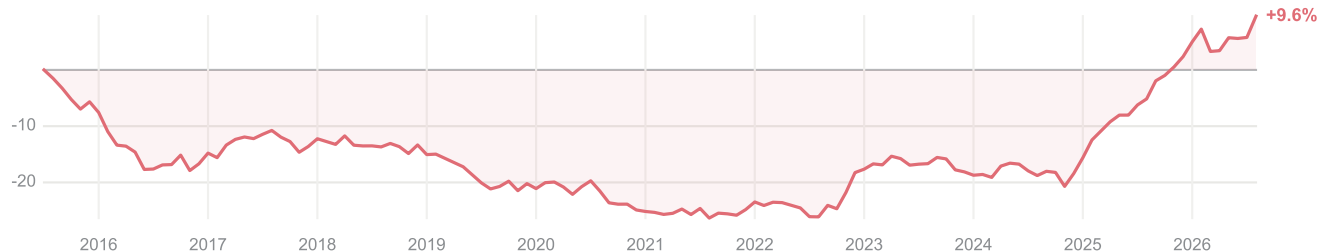
Historical Growth

CAGR +9.9%

**018 ● Gross Profit Margin**

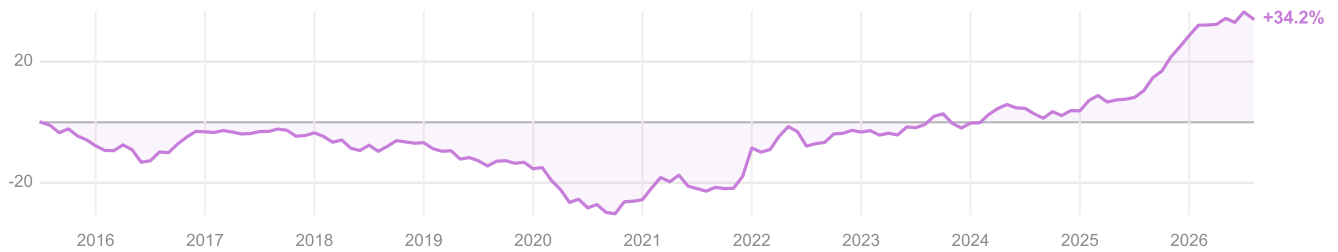
Earnings Quality

CAGR +9.9%

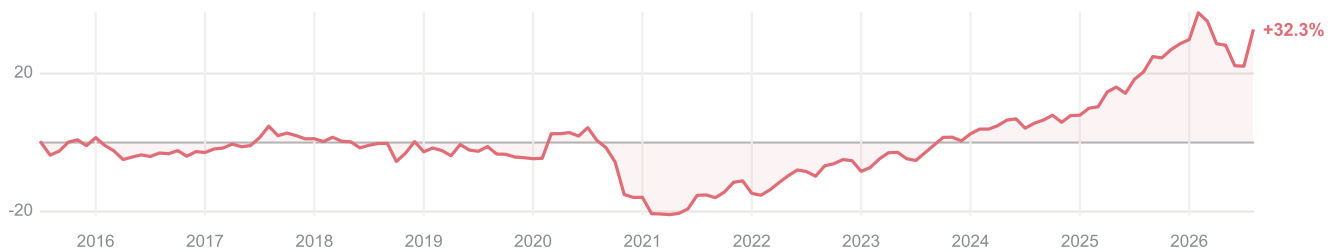


019 ● Log of Total Last Quarter Assets

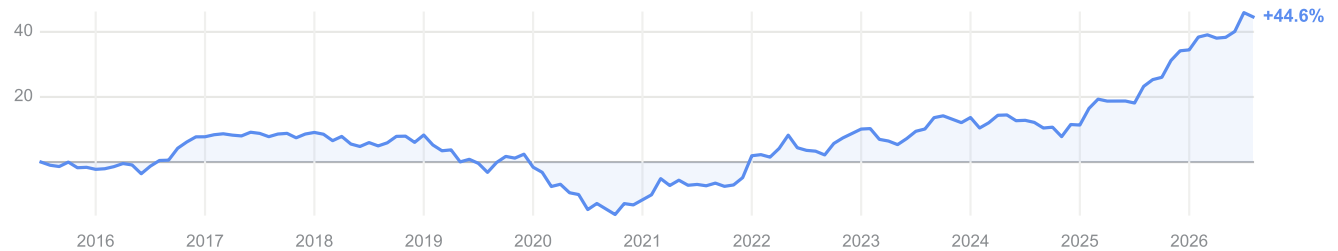
Size CAGR +9.7%

**020 ● Trailing 12-Month Earnings/Trailing 12-Month Sales**

Earnings Quality CAGR +9.4%

**021 ● Operating Earnings to Price Ratio**

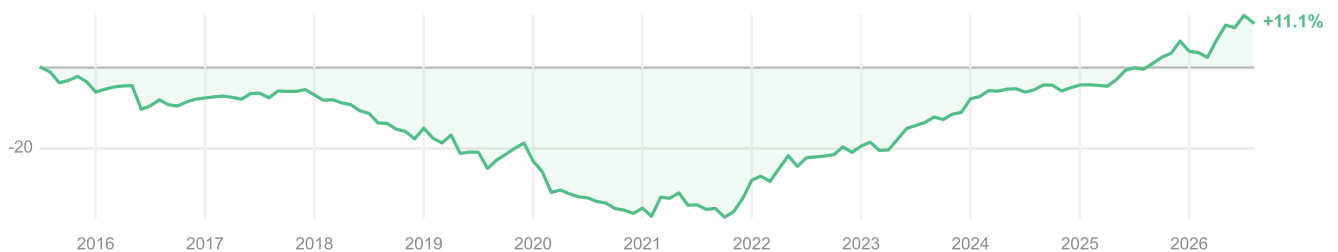
Valuation CAGR +9.2%

**022 ● 3Y Chg in Operating Cash Flow to Price**

Historical Growth CAGR +9.1%

**023 ● Adjusted Inverse PEG**

Analyst Expectations CAGR +9.1%

**024 ● 3M Revision in FY1 EPS Estimate**

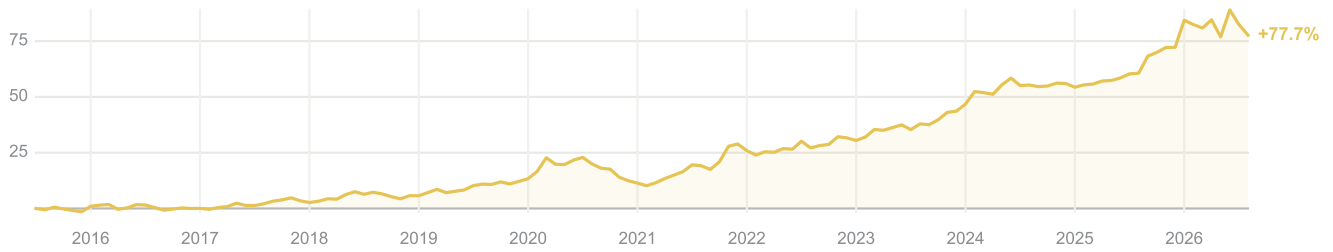
Analyst Expectations CAGR +8.8%



025 ● **Return on Assets**

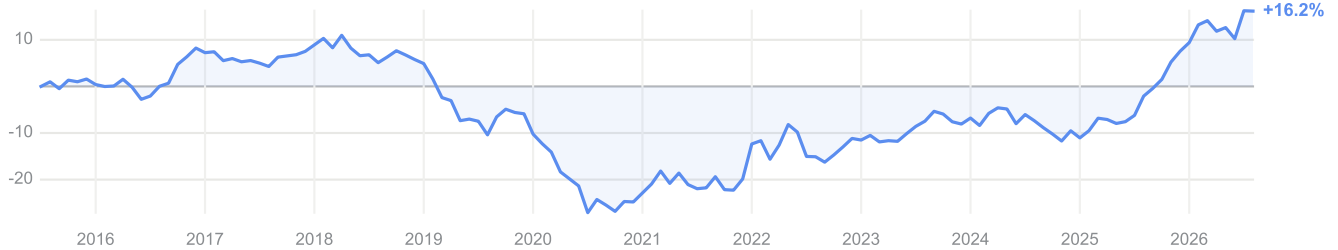
Capital Efficiency

CAGR +8.8%

026 ● **Sales to EV Ratio**

Valuation

CAGR +8.8%

027 ● **Inverse PEG**

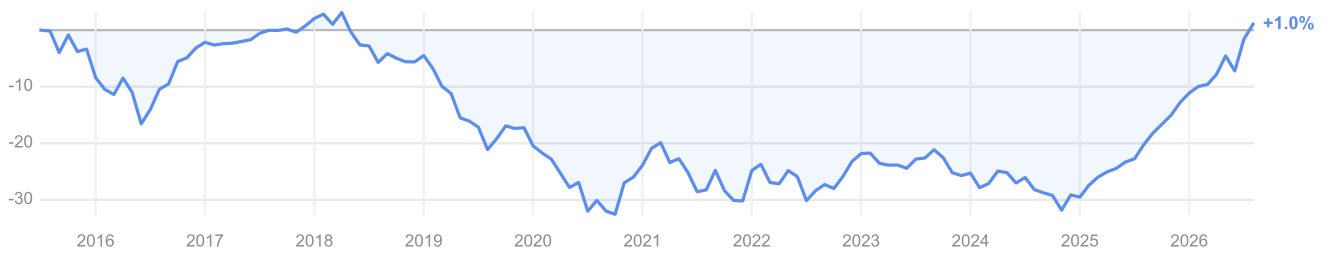
Analyst Expectations

CAGR +8.7%

028 ● **Cash to Price**

Valuation

CAGR +8.7%

029 ● **60M CAPM Beta**

Volatility

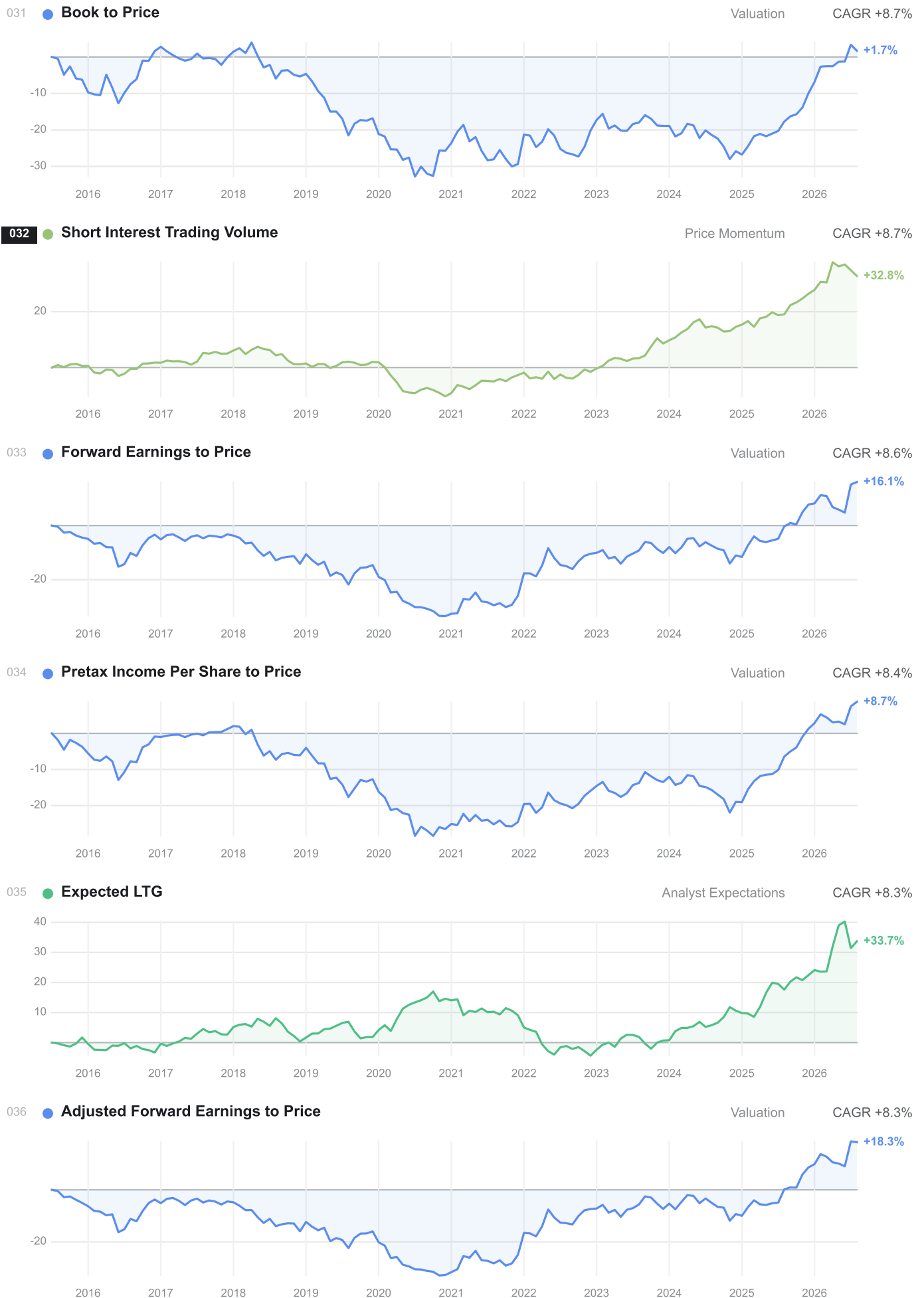
CAGR +8.7%

030 ● **Earnings to Price**

Valuation

CAGR +8.7%

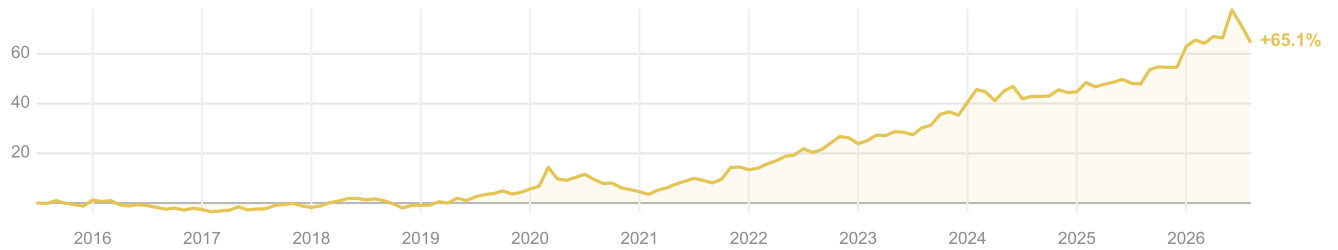




037 ● **Return on Invested Capital**

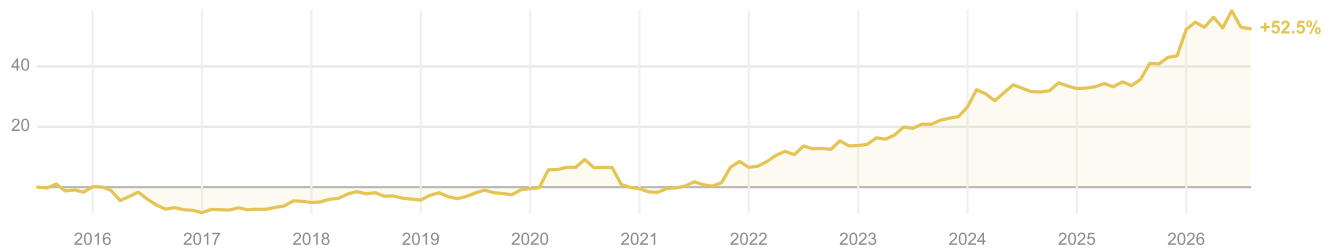
Capital Efficiency

CAGR +8.1%

038 ● **EBITDA to Assets**

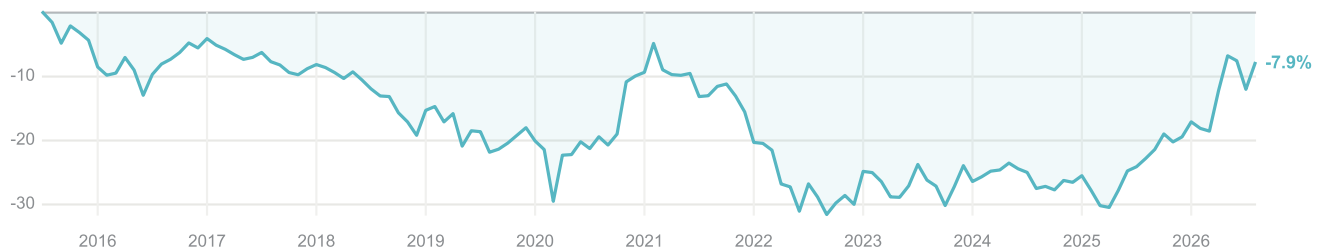
Capital Efficiency

CAGR +7.9%

039 ● **1M Realized Price Volatility**

Volatility

CAGR +7.9%

040 ● **Return on Equity**

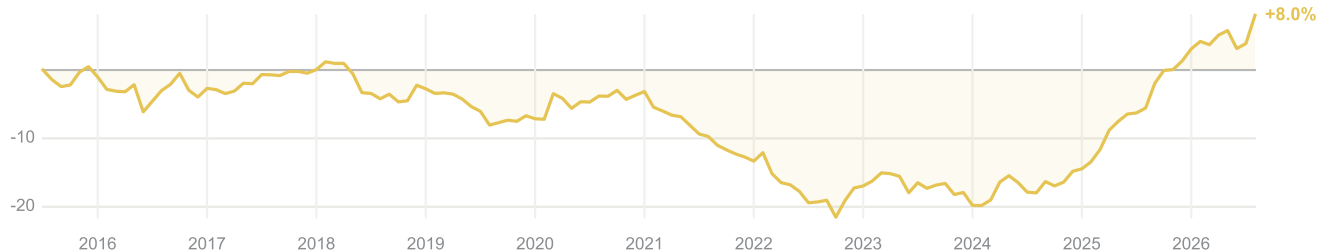
Capital Efficiency

CAGR +7.9%

041 ● **Long Term Debt to Assets Ratio**

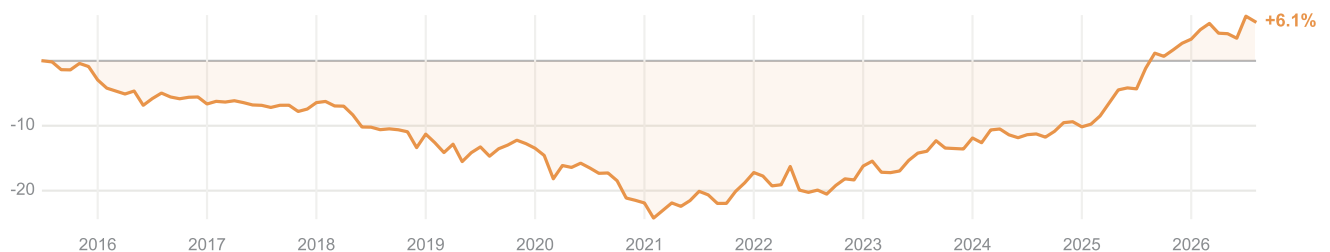
Capital Efficiency

CAGR +7.8%

042 ● **1Y Chg in Sales to Price**

Historical Growth

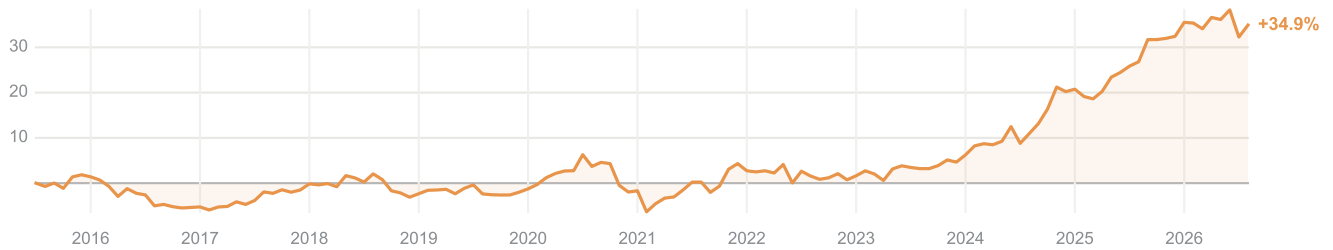
CAGR +7.7%



043 1Y Chg in Asset Adjusted EPS

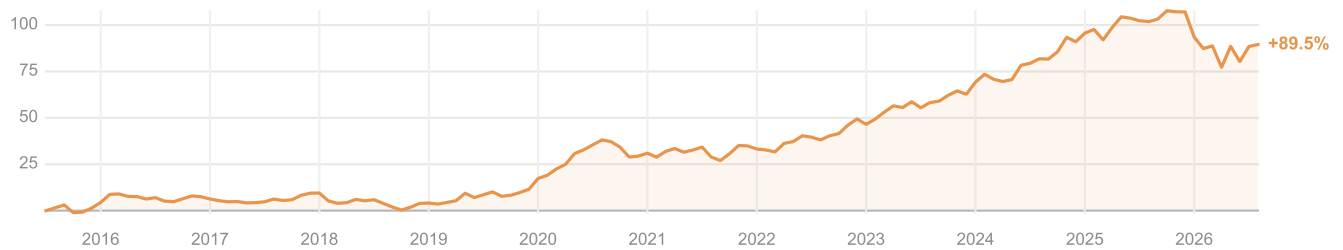
Historical Growth

CAGR +7.7%

**044 Consecutive Quarters of Declines in TTM Overhead/Sales**

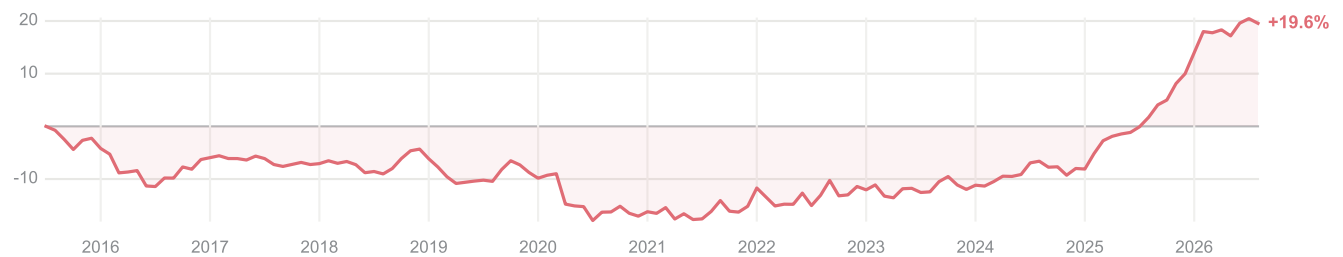
Historical Growth

CAGR +7.7%

**045 ROA 20 Qtr Standard Deviation**

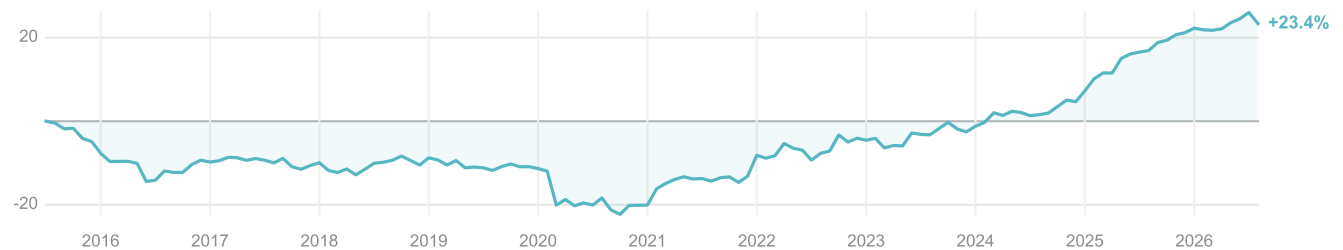
Earnings Quality

CAGR +7.7%

**046 Book Leverage**

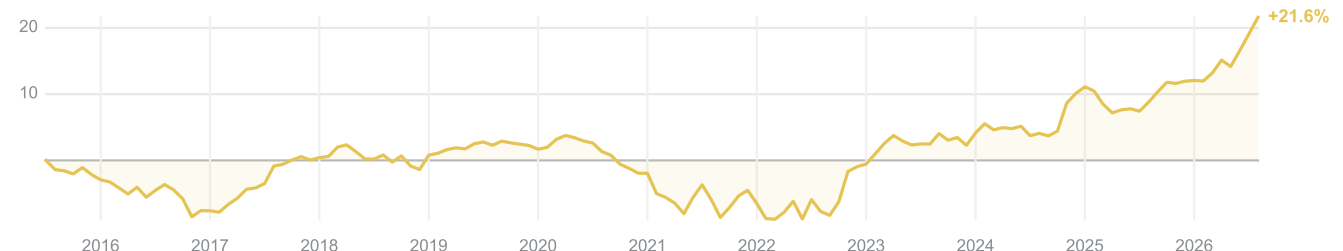
Volatility

CAGR +7.6%

**047 1Y Chg in Stock Buybacks**

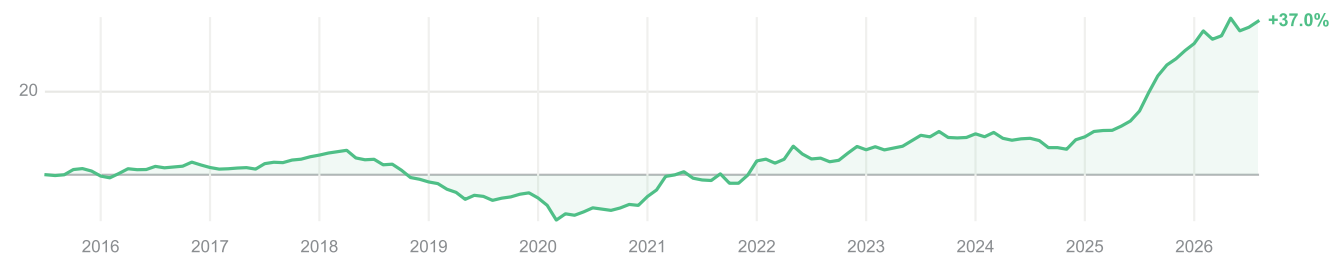
Capital Efficiency

CAGR +7.6%

**048 Earnings Surprise**

Analyst Expectations

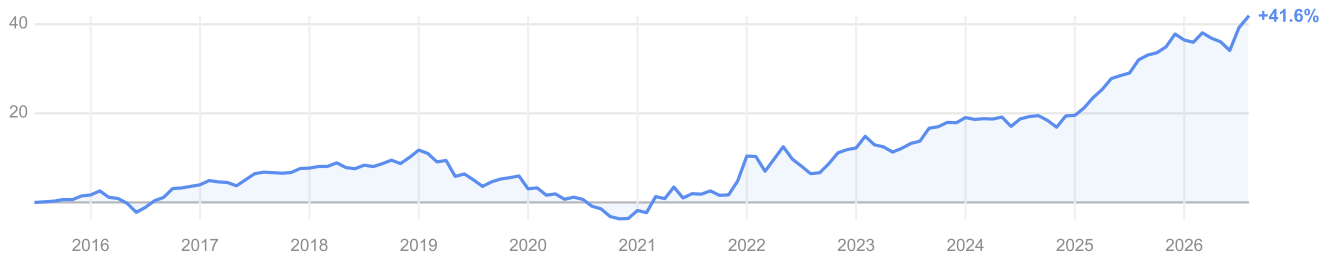
CAGR +7.5%



049 ● Free Cash Flow to Enterprise Value

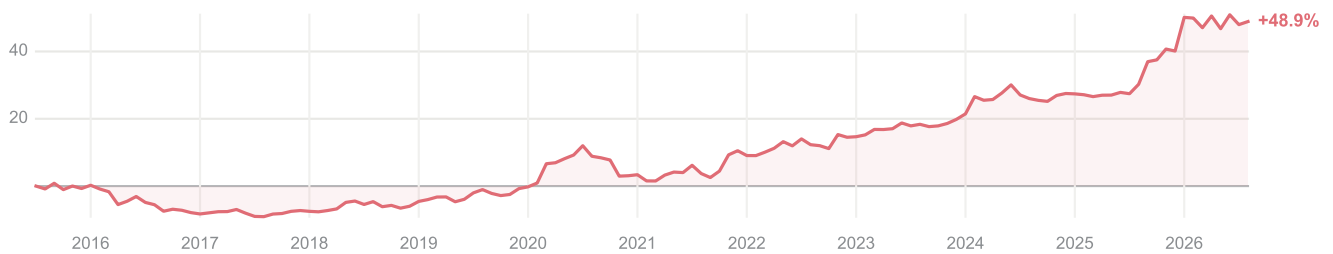
Valuation

CAGR +7.5%

**050 ● Cash Flow to Total Asset**

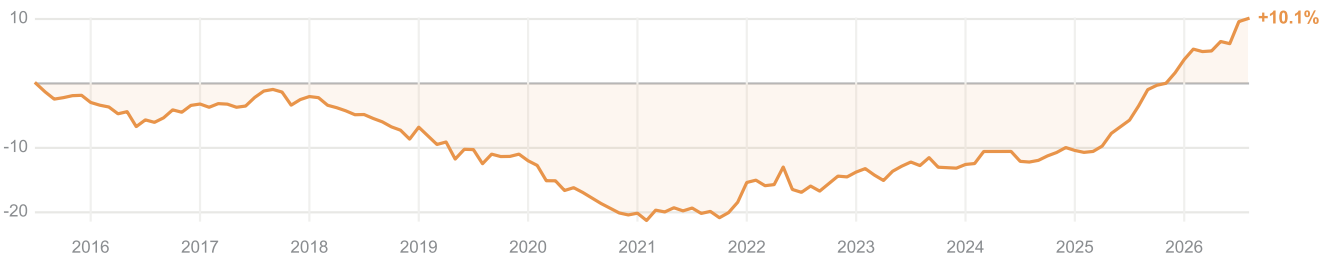
Earnings Quality

CAGR +7.5%

**051 ● 1Y Chg in Cash Flow to Price**

Historical Growth

CAGR +7.4%

**052 ● Adj 1Y Chg in Earnings per Share**

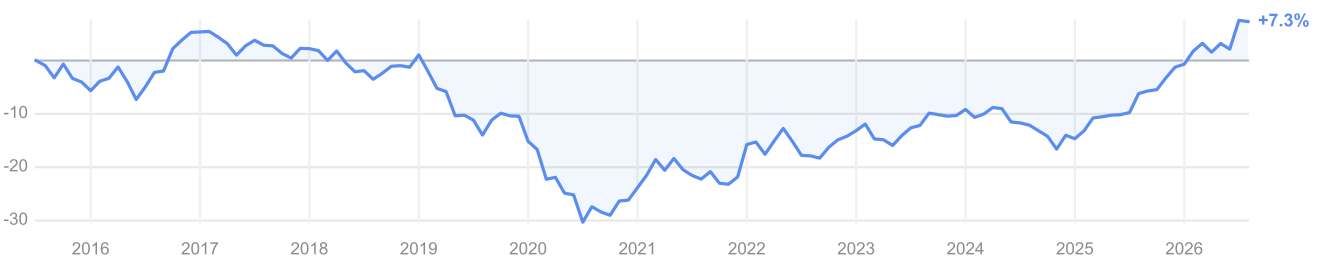
Historical Growth

CAGR +7.4%

**053 ● Operating Cash Flow to Price**

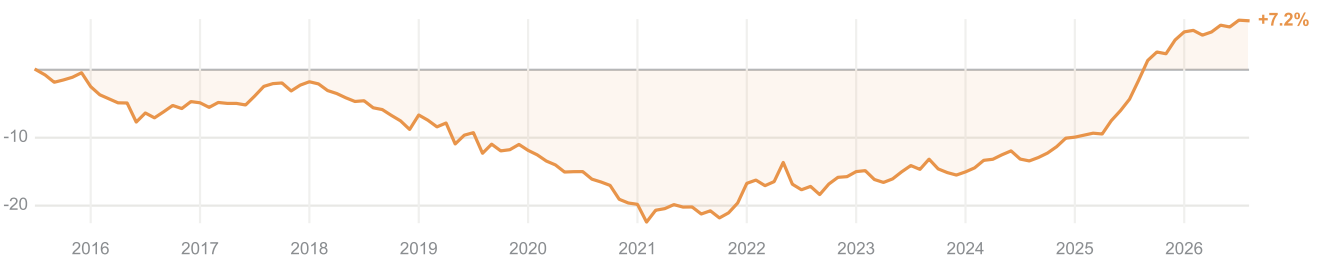
Valuation

CAGR +7.3%

**054 ● 1Y Chg in Earnings to Price**

Historical Growth

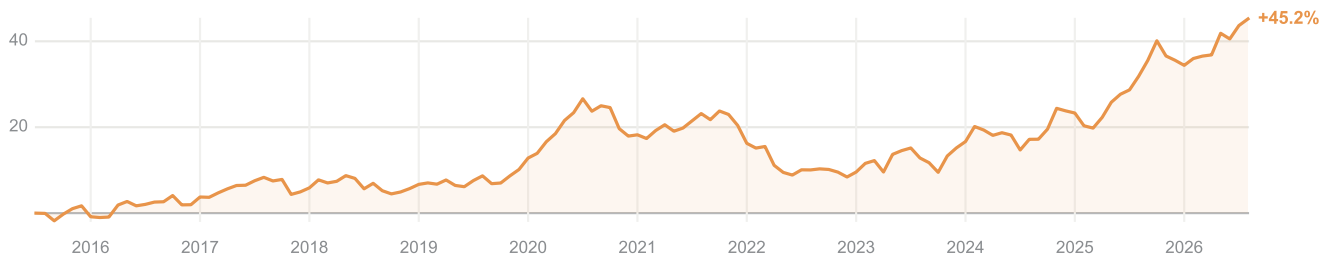
CAGR +7.3%



055 ● **3Y Chg in Operating Cash Flow per Share**

Historical Growth

CAGR +7.3%

056 ● **90 Day Coefficient of Variation**

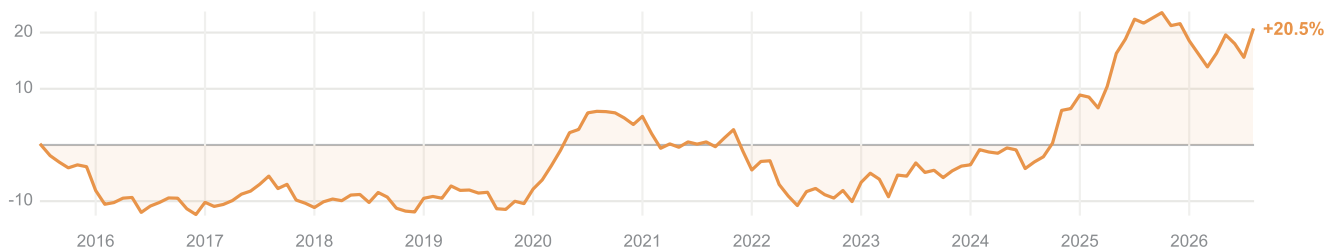
Volatility

CAGR +7.3%

057 ● **12-Quarter Trendline in Trailing 12-Month Cash Flows**

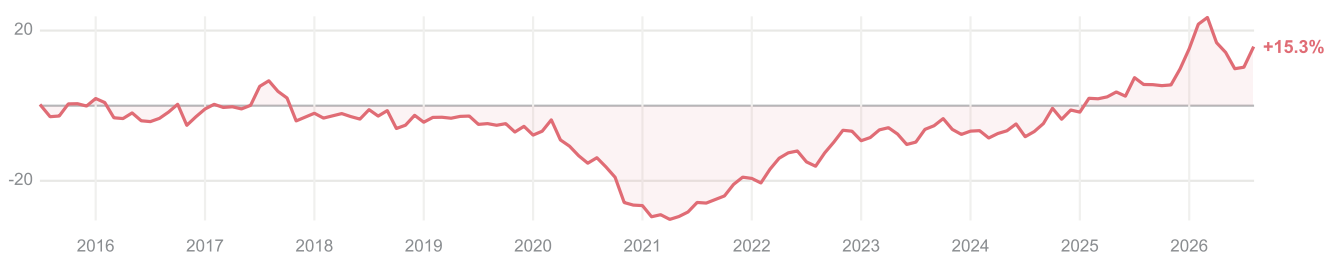
Historical Growth

CAGR +7.3%

058 ● **EBITDA Margin**

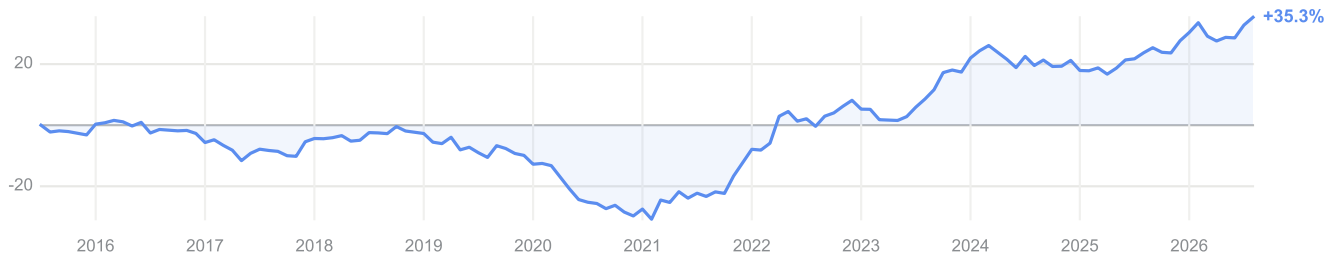
Earnings Quality

CAGR +7.3%

059 ● **Adj Free Cash Flow to Price**

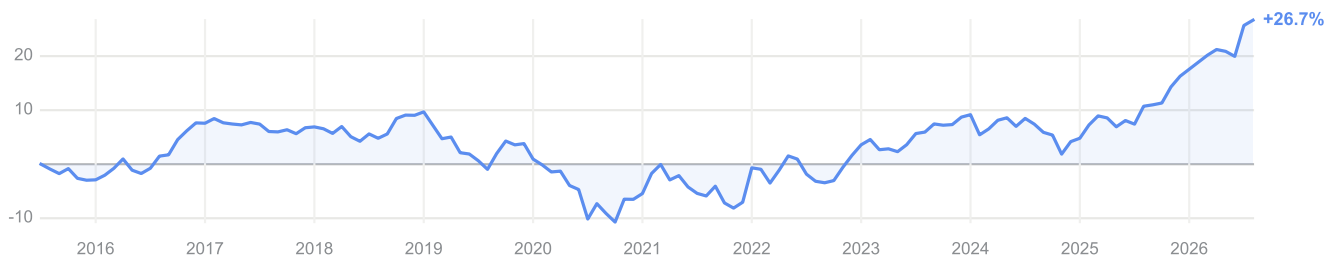
Valuation

CAGR +7.2%

060 ● **Normalized EP**

Valuation

CAGR +7.2%



061 ● Working Capital to Sales

Earnings Quality

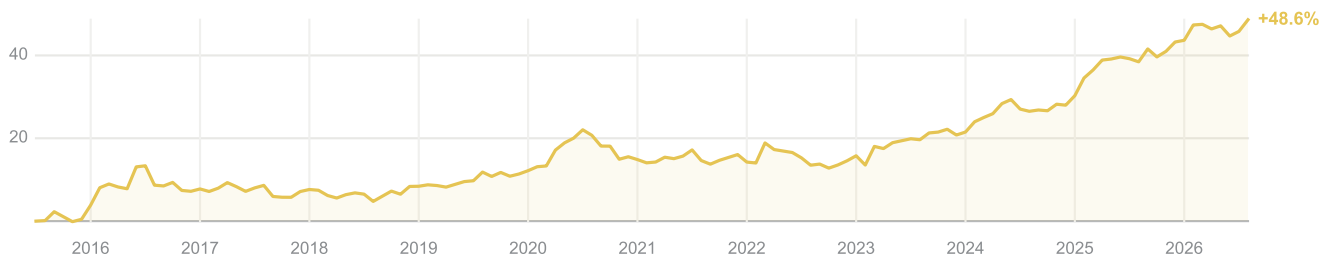
CAGR +7.2%



062 ● Receivables Turnover Ratio

Capital Efficiency

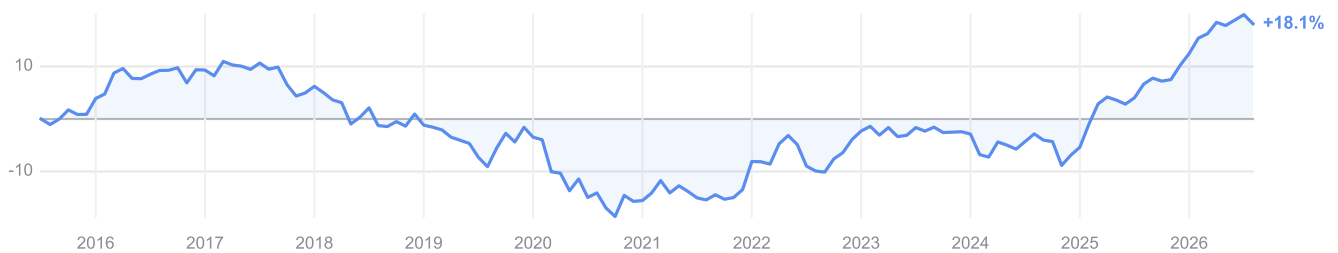
CAGR +7.2%



063 ● Dividends to Price Ratio

Valuation

CAGR +7.2%



064 ● Sales to Invested Capital

Earnings Quality

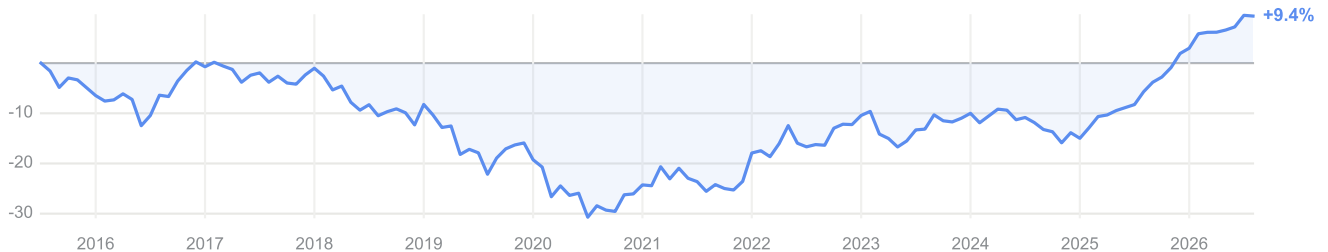
CAGR +7.2%



065 ● Adjusted EBIT to Price

Valuation

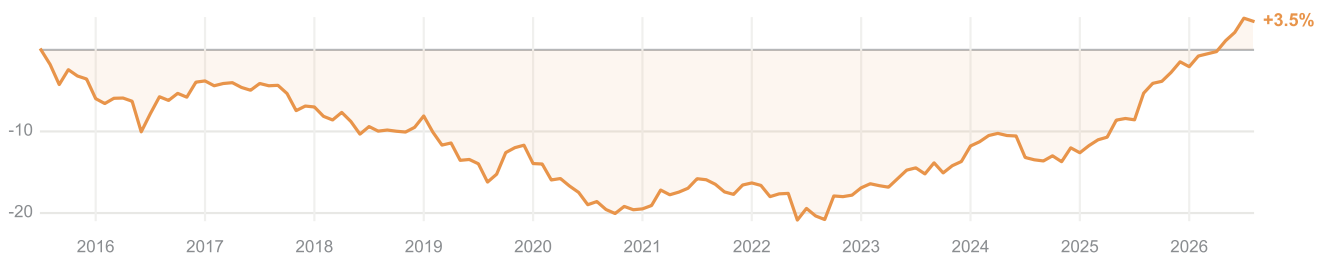
CAGR +7.1%



066 ● 3Y Chg in Cash Flow to Price

Historical Growth

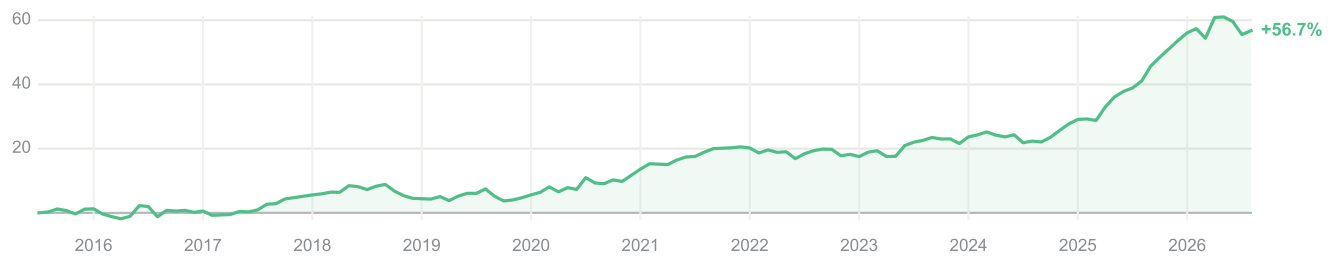
CAGR +7.1%



067 ● Street Revision Magnitude

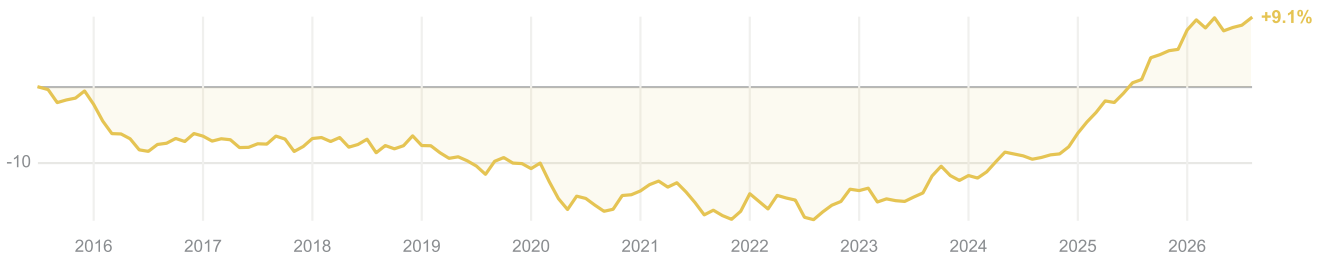
Analyst Expectations

CAGR +7.1%

**068 ● Capital Expenditure to Total Assets**

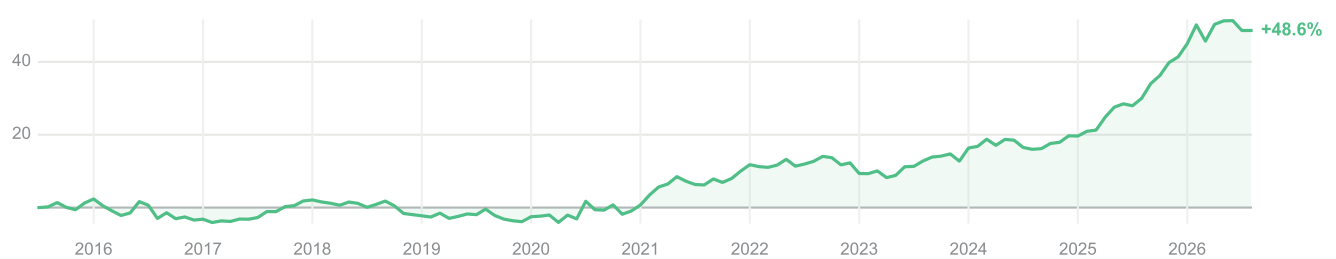
Capital Efficiency

CAGR +7.0%

**069 ● 3M Revision in FY2 EPS Estimate**

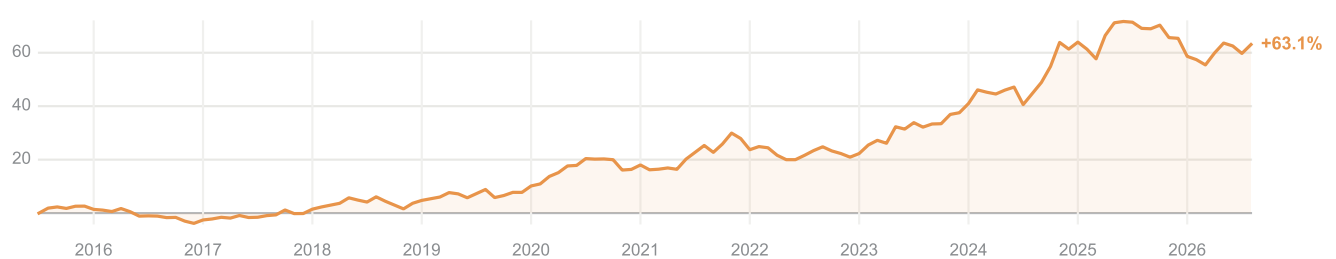
Analyst Expectations

CAGR +7.0%

**070 ● Adjusted 3Y Chg in Asset Adjusted Free Cash Flow**

Historical Growth

CAGR +7.0%

**071 ● Analyst Dispersion for FY1 EPS**

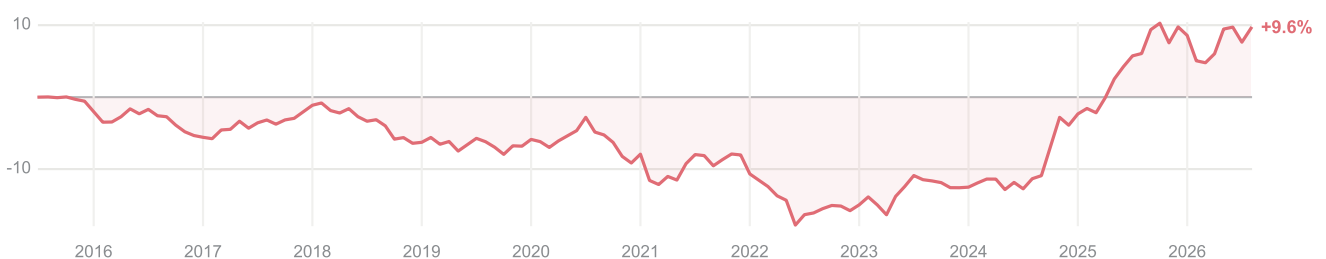
Analyst Expectations

CAGR +7.0%

**072 ● Year over Year Change in SGA to Sales**

Earnings Quality

CAGR +6.9%



073 ● 1Y Chg in Asset Adjusted Cash Flow

Historical Growth

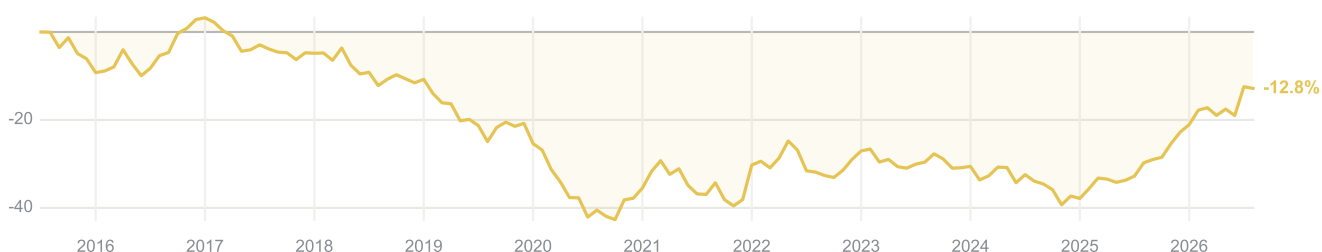
CAGR +6.9%



074 ● Tobin's Q Ratio

Capital Efficiency

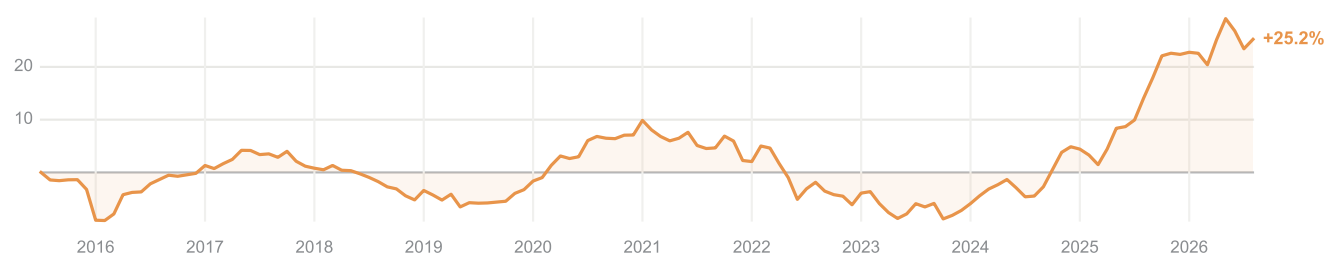
CAGR +6.8%



075 ● 12-Quarter Trendline in Trailing 12-Month Earnings

Historical Growth

CAGR +6.8%



076 ● Expected Earnings Growth: Fiscal Year 2/Fiscal Year 1

Historical Growth

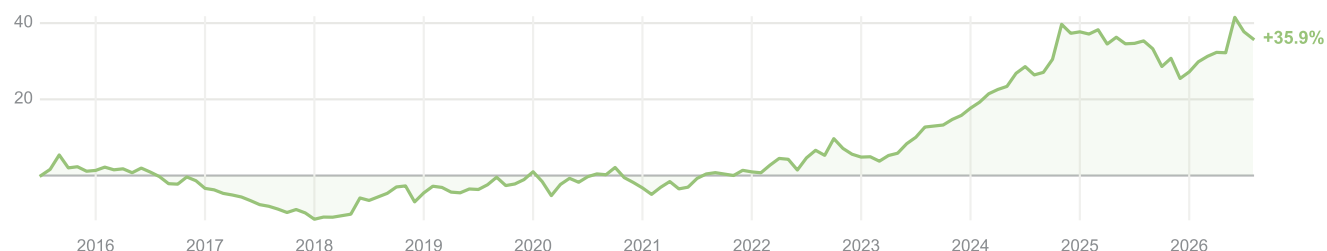
CAGR +6.8%



077 ● Institution Ownership Level - Active Manager

Price Momentum

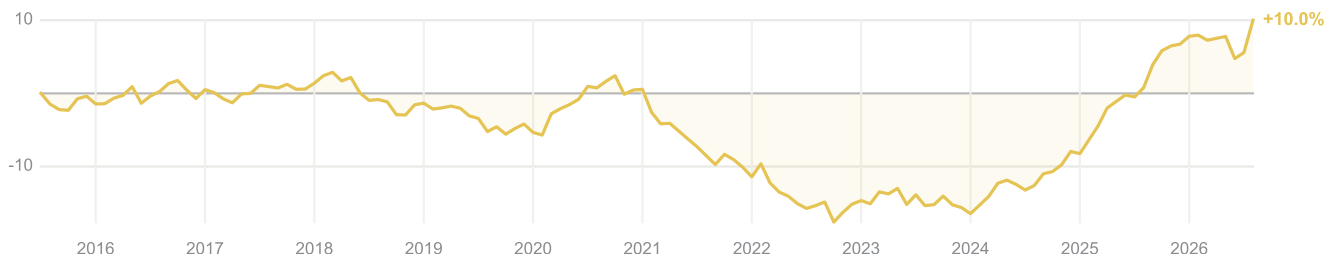
CAGR +6.8%



078 ● Debt to Assets Ratio

Capital Efficiency

CAGR +6.7%



079 ● **Stability of Change in Number of Holders**

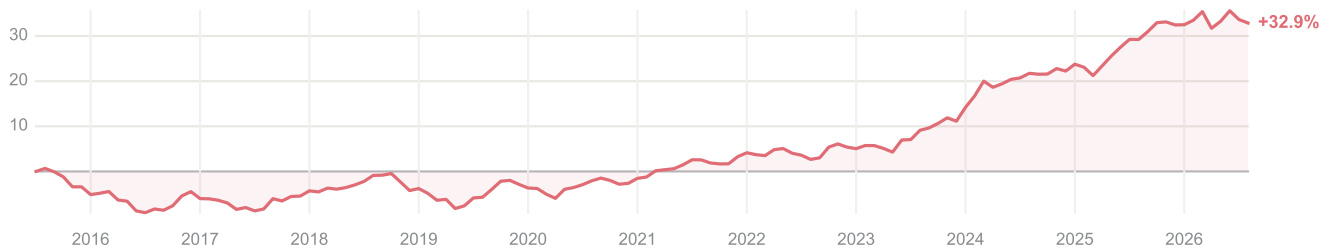
Price Momentum

CAGR +6.7%

080 ● **Equity Turnover**

Earnings Quality

CAGR +6.7%

081 ● **Operating Cash Flow to Asset**

Earnings Quality

CAGR +6.6%

082 ● **Operating Cash Flow to Invested Capital**

Earnings Quality

CAGR +6.6%

083 ● **Change EPS to Sales**

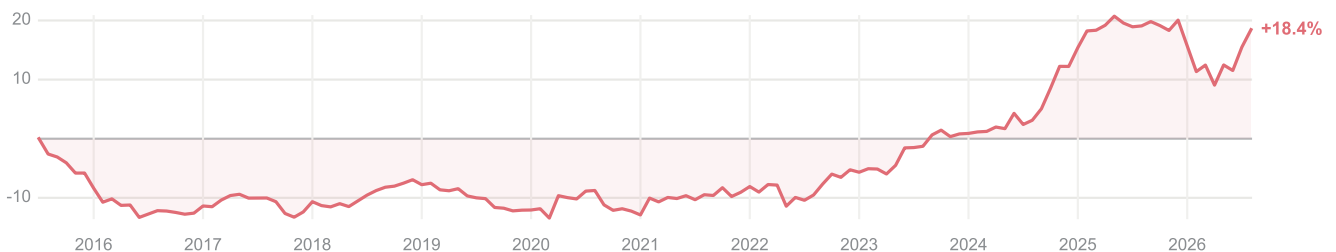
Earnings Quality

CAGR +6.6%

084 ● **Balance Sheet Accruals**

Earnings Quality

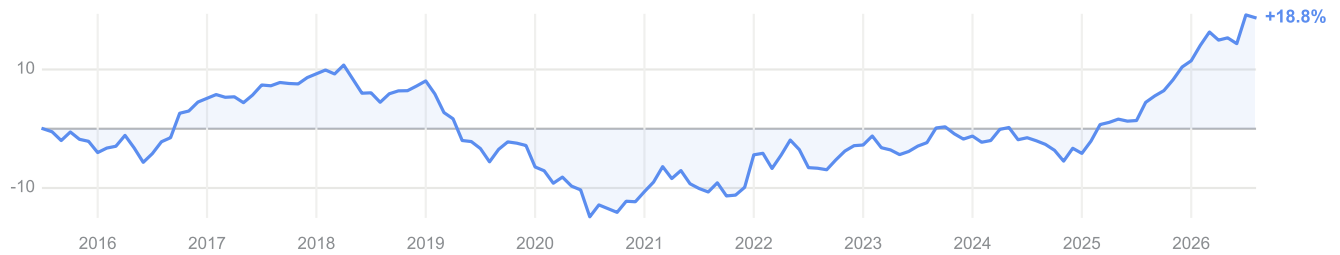
CAGR +6.6%



085 ● Operating Cash Flow to Enterprise Value

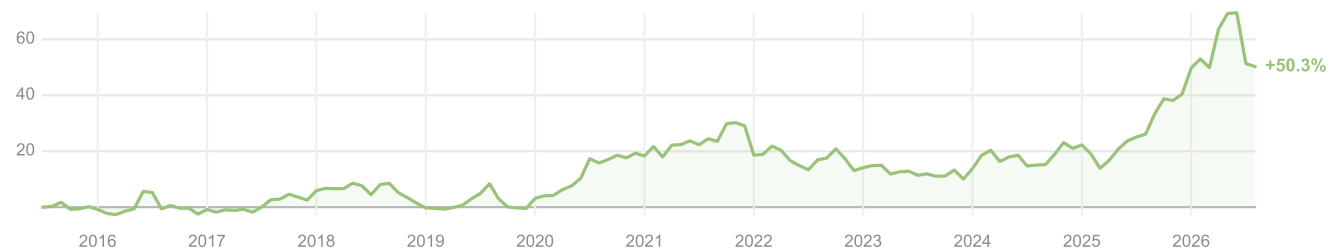
Valuation

CAGR +6.5%

**086 ● Percent Above 260-Day Low**

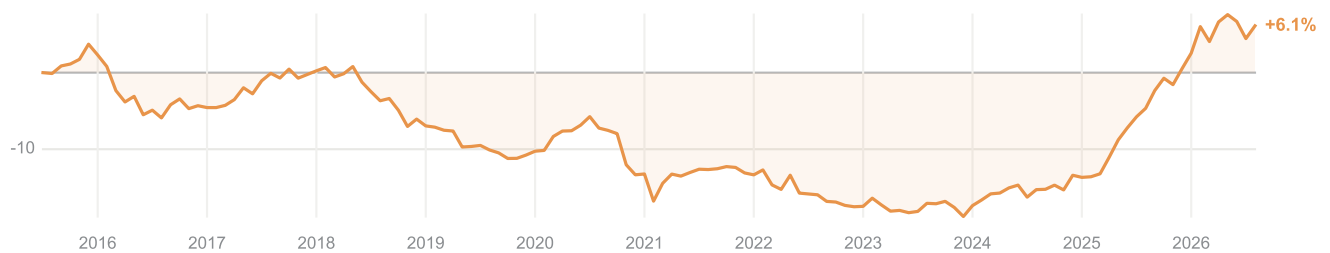
Price Momentum

CAGR +6.5%

**087 ● 1Y Chg in Operating Margin**

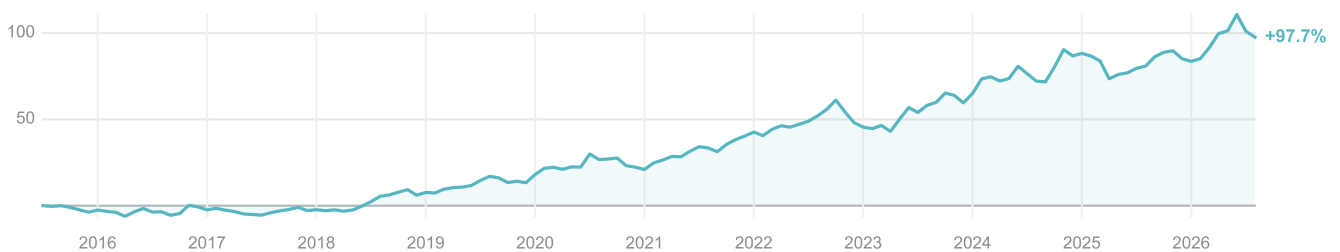
Historical Growth

CAGR +6.4%

**088 ● Amihud Illiquidity Measure**

Volatility

CAGR +6.3%

**089 ● EBITDA to Price**

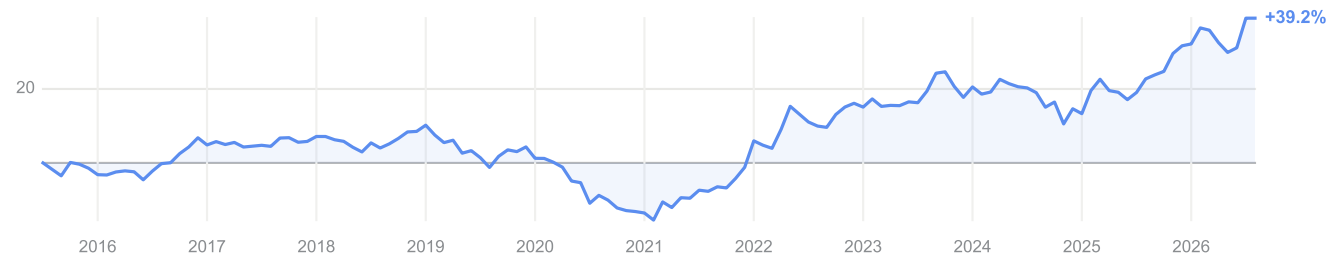
Valuation

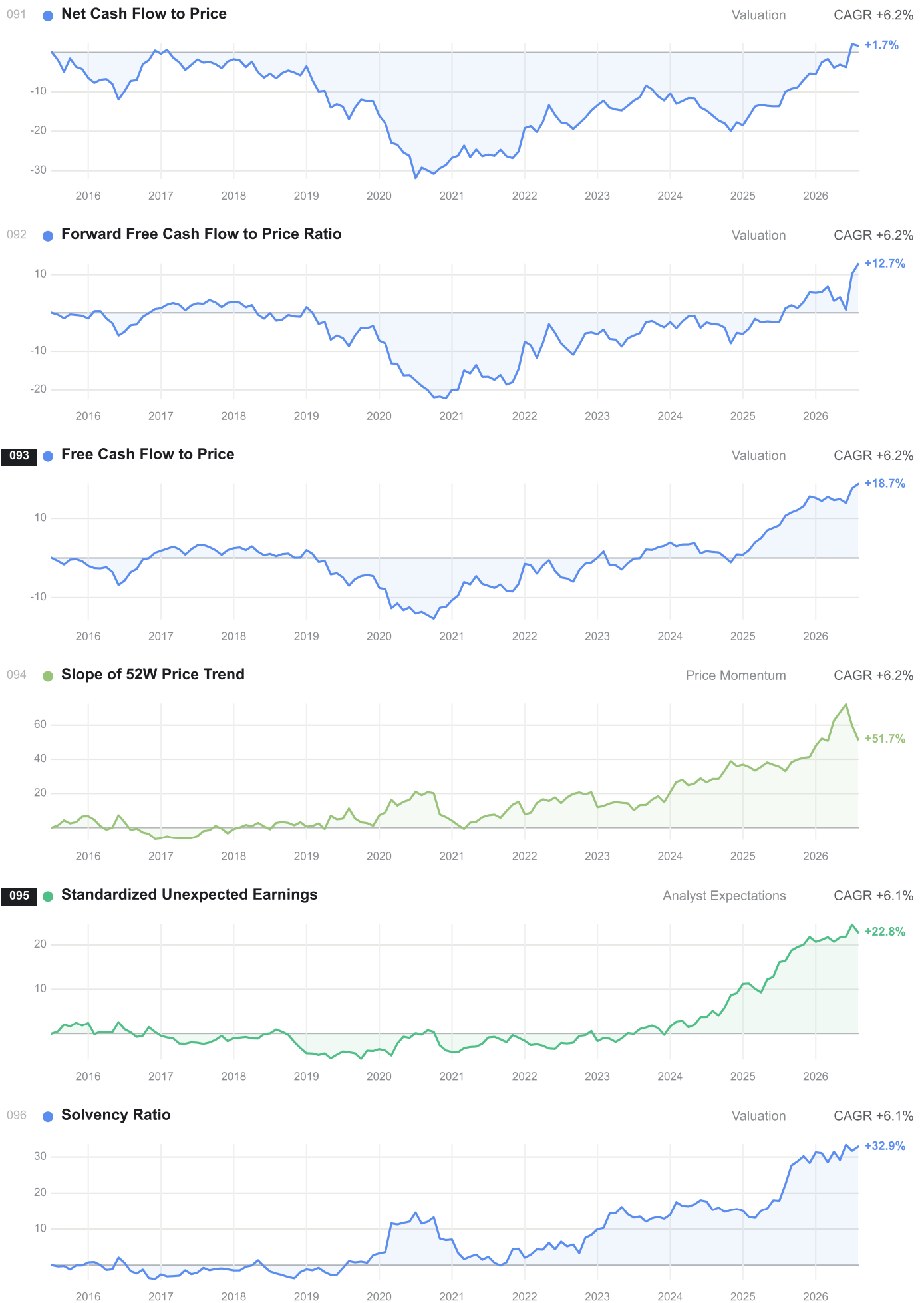
CAGR +6.3%

**090 ● Growth Flow to Price**

Valuation

CAGR +6.3%

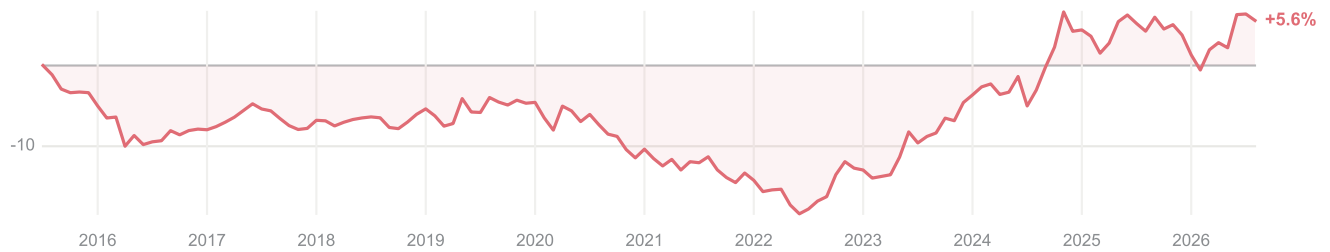




097 ● **Inventory Turnover**

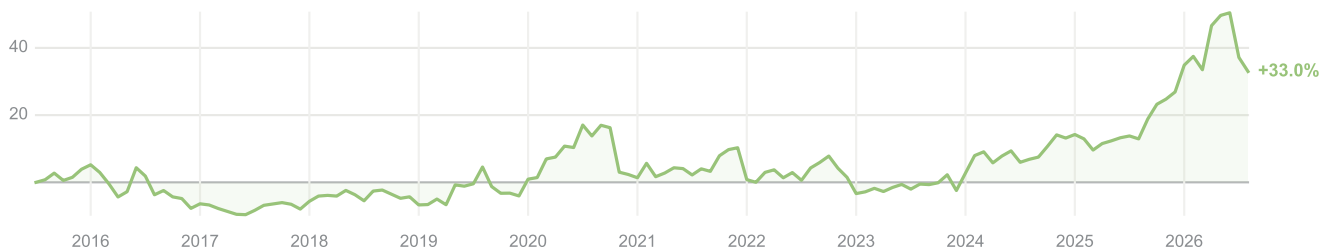
Earnings Quality

CAGR +6.1%

098 ● **39W Lagged Return**

Price Momentum

CAGR +6.0%

099 ● **Capital Expenditure to Sales**

Earnings Quality

CAGR +6.0%

100 ● **Operating Margin**

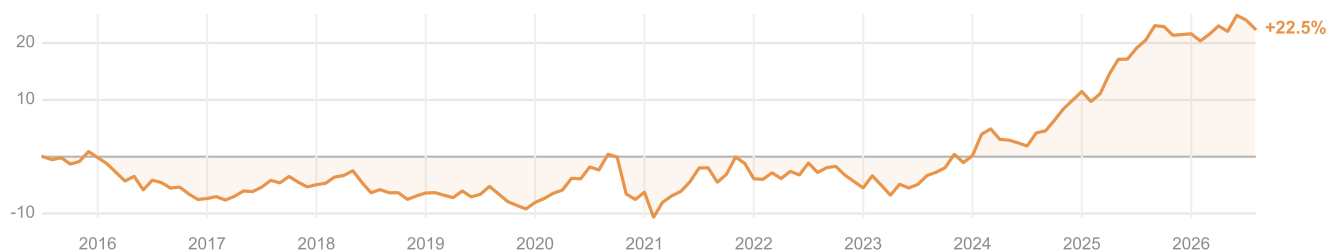
Earnings Quality

CAGR +6.0%

101 ● **1Y Chg in Operating Cash Flow per Share**

Historical Growth

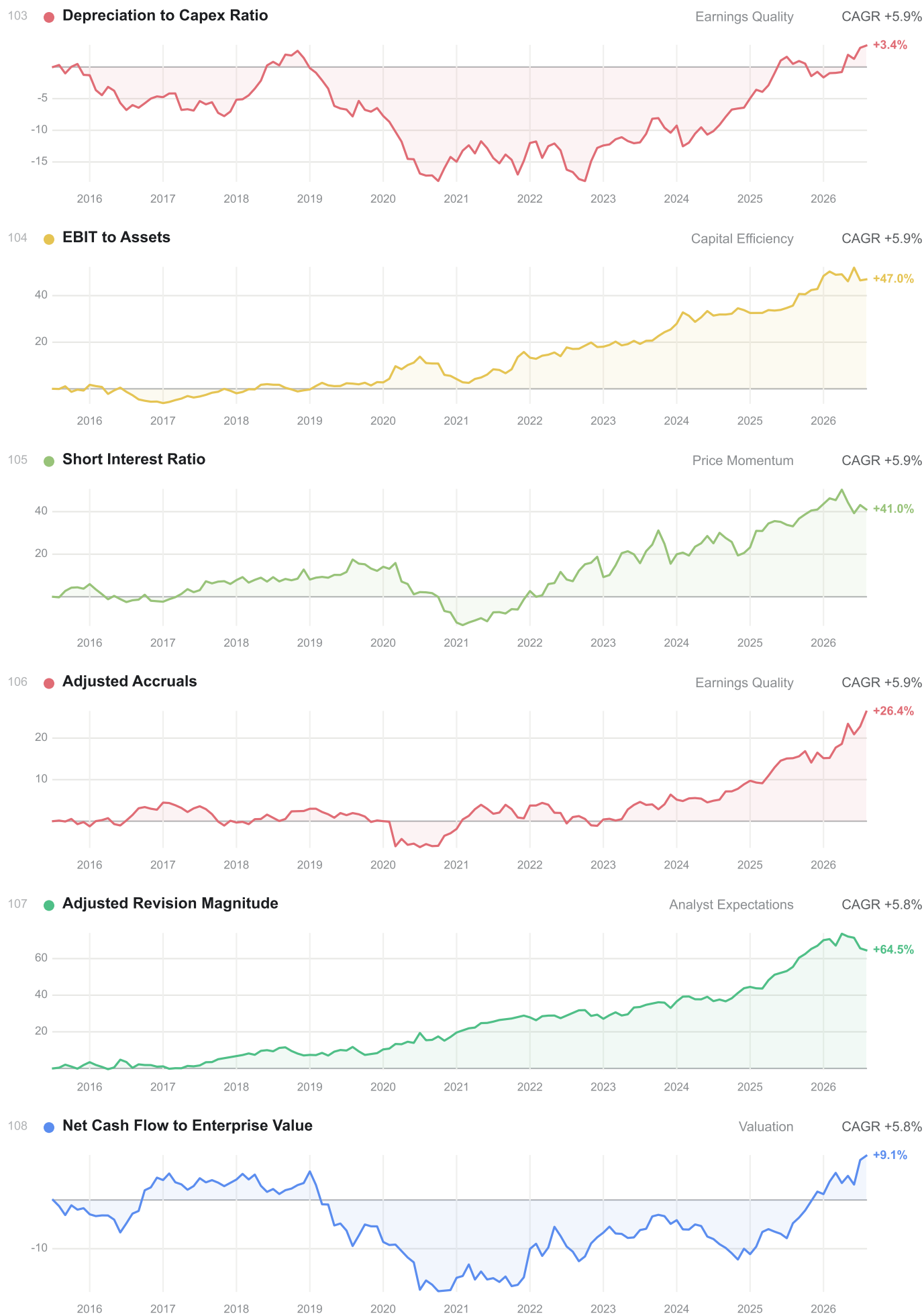
CAGR +5.9%

102 ● **Cash Ratio**

Earnings Quality

CAGR +5.9%

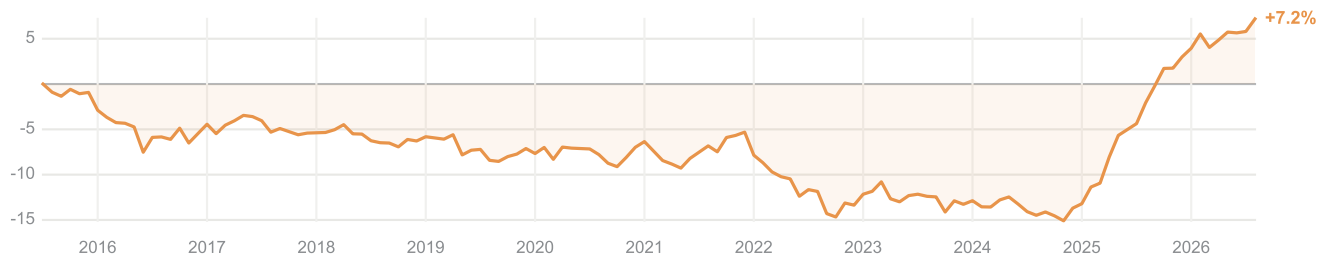




109 ● Dividend Growth

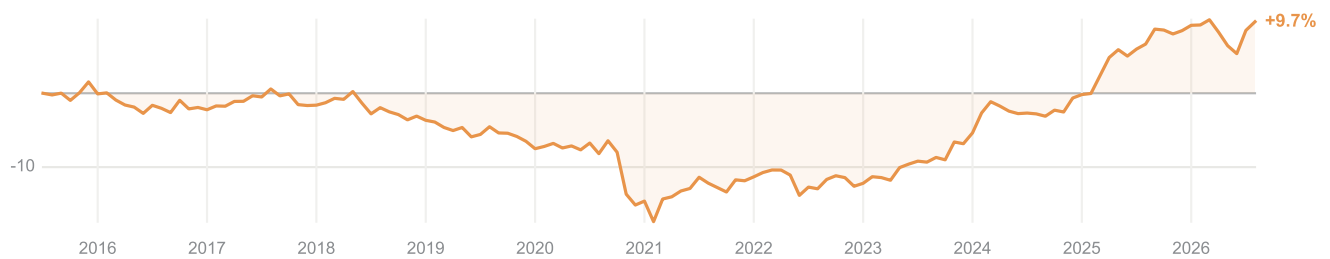
Historical Growth

CAGR +5.8%

**110 ● 1Y Chg in Operating Cash Flow to Price**

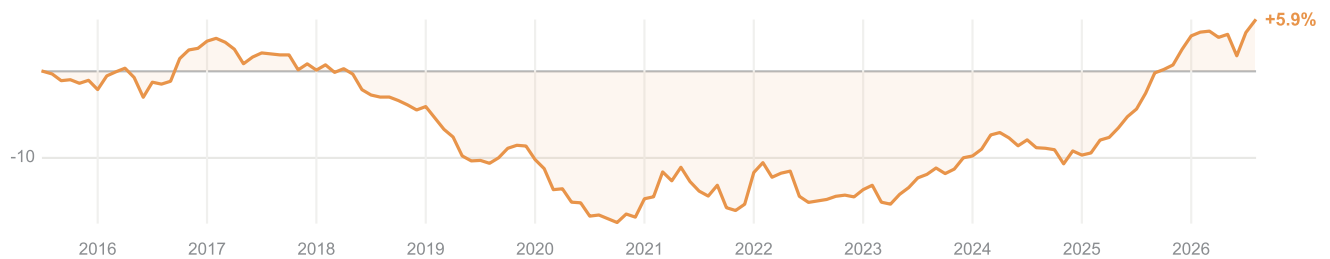
Historical Growth

CAGR +5.7%

**111 ● 1Y Chg in Free Cash Flow to Price**

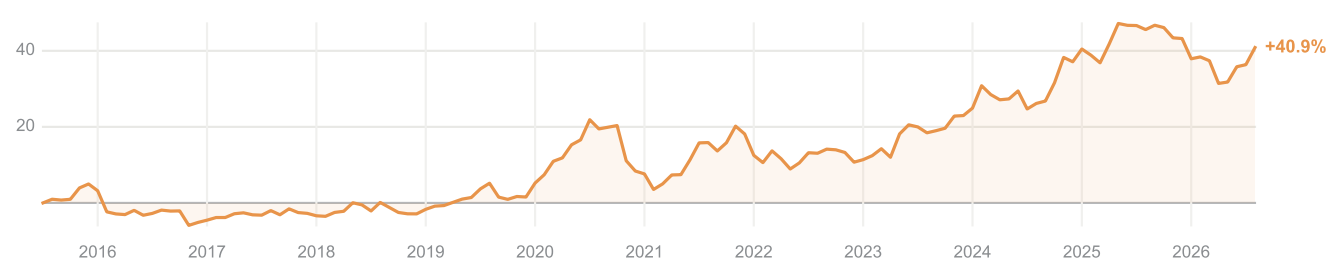
Historical Growth

CAGR +5.7%

**112 ● 3Y Chg in Asset Adjusted Operating Cash Flow**

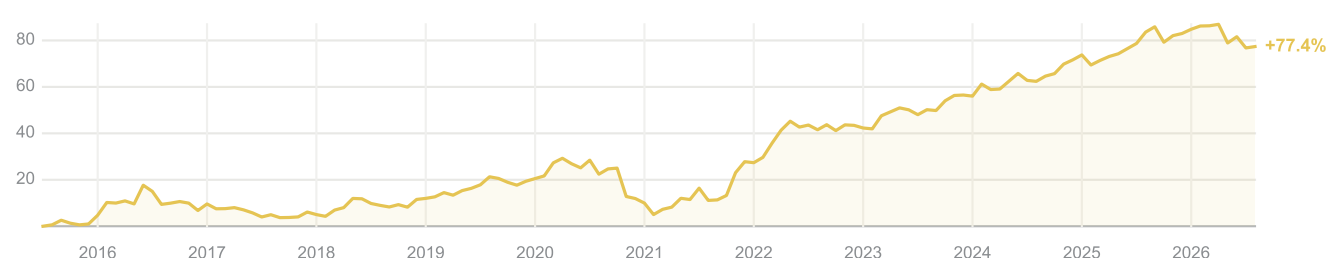
Historical Growth

CAGR +5.7%

**113 ● Operating Cash Flow Ratio**

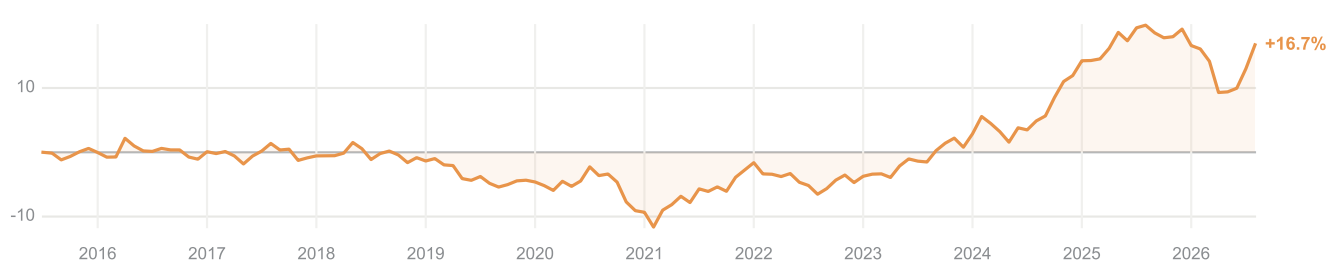
Capital Efficiency

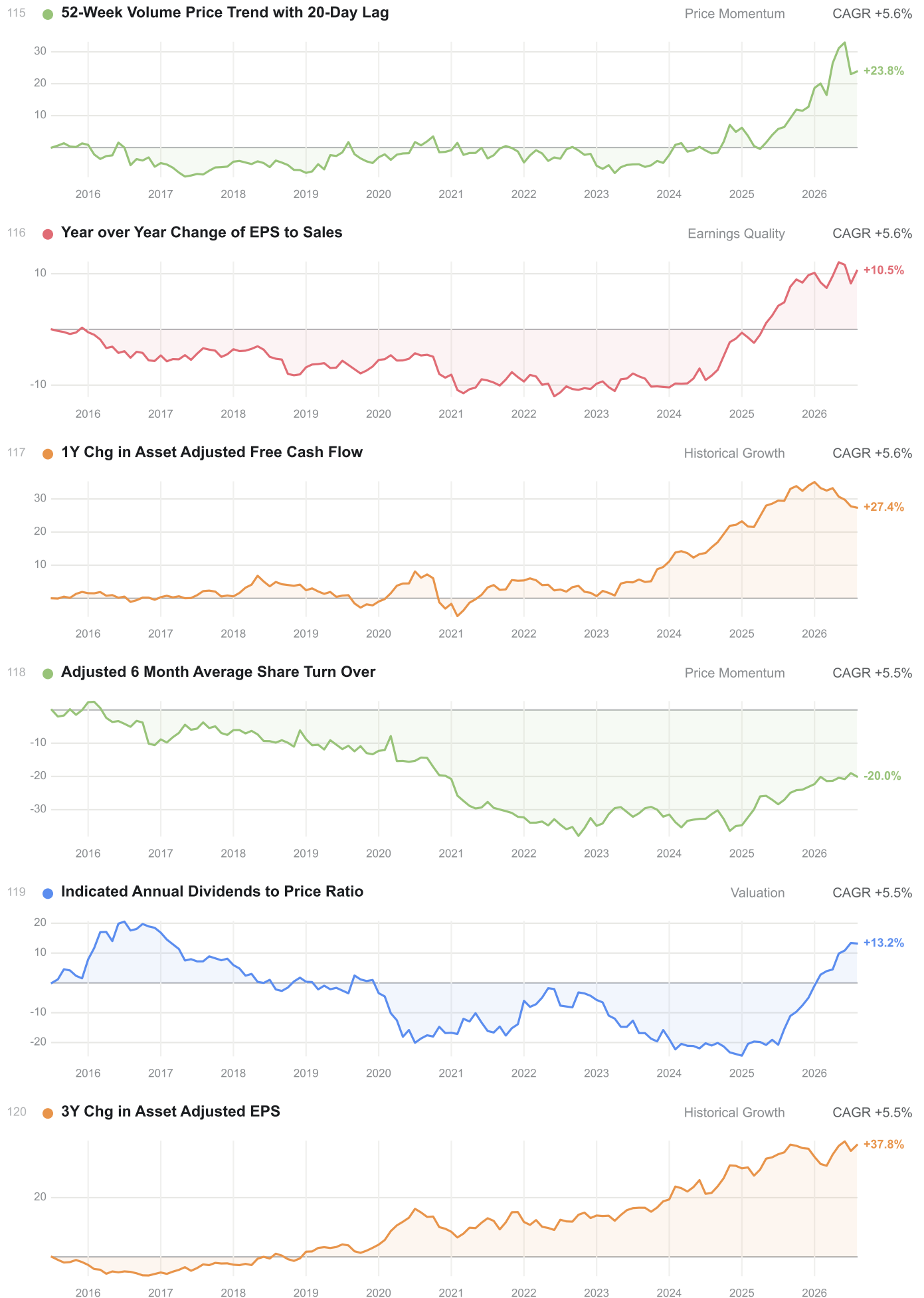
CAGR +5.6%

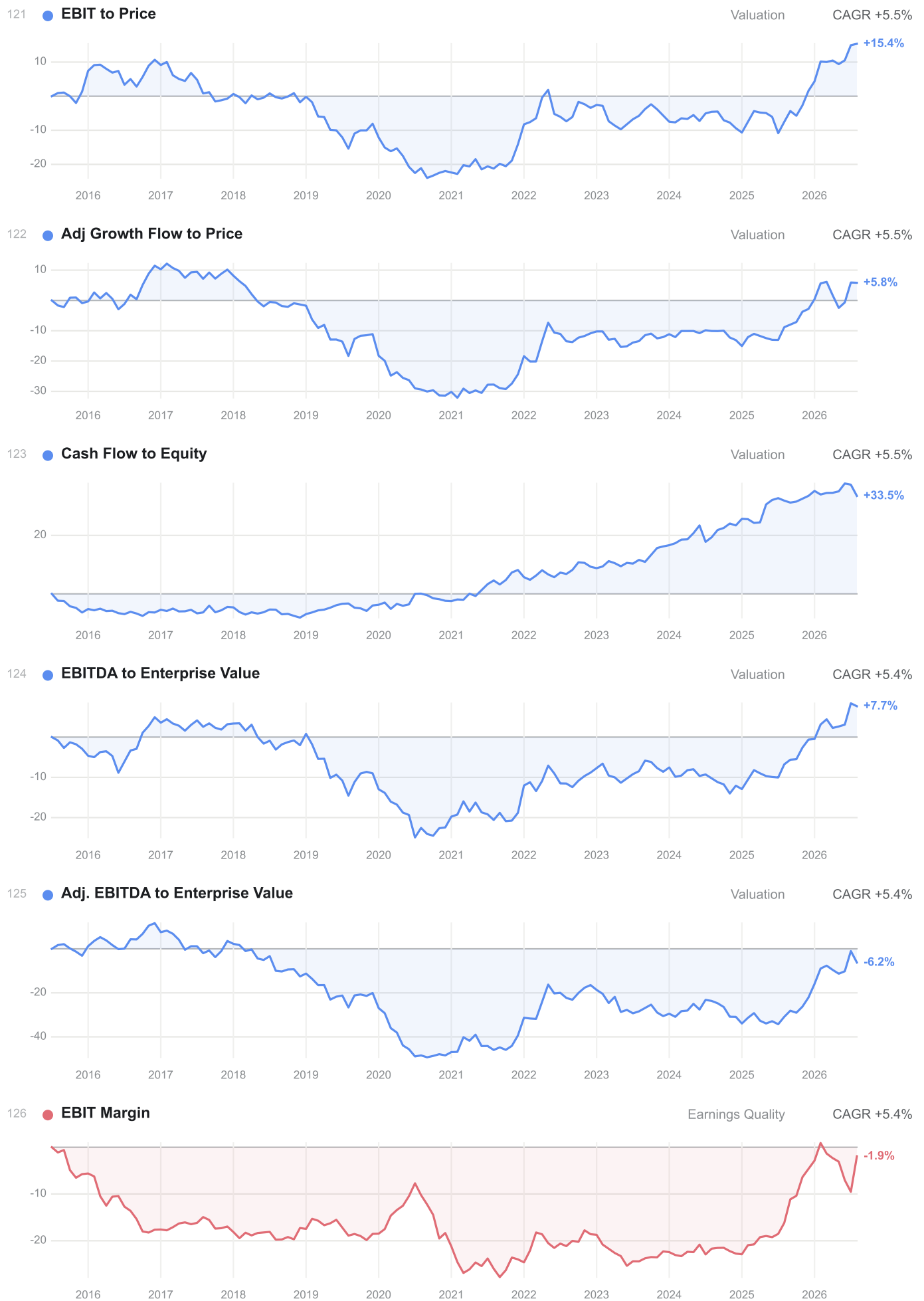
**114 ● 3Y Chg in Free Cash Flow to Price**

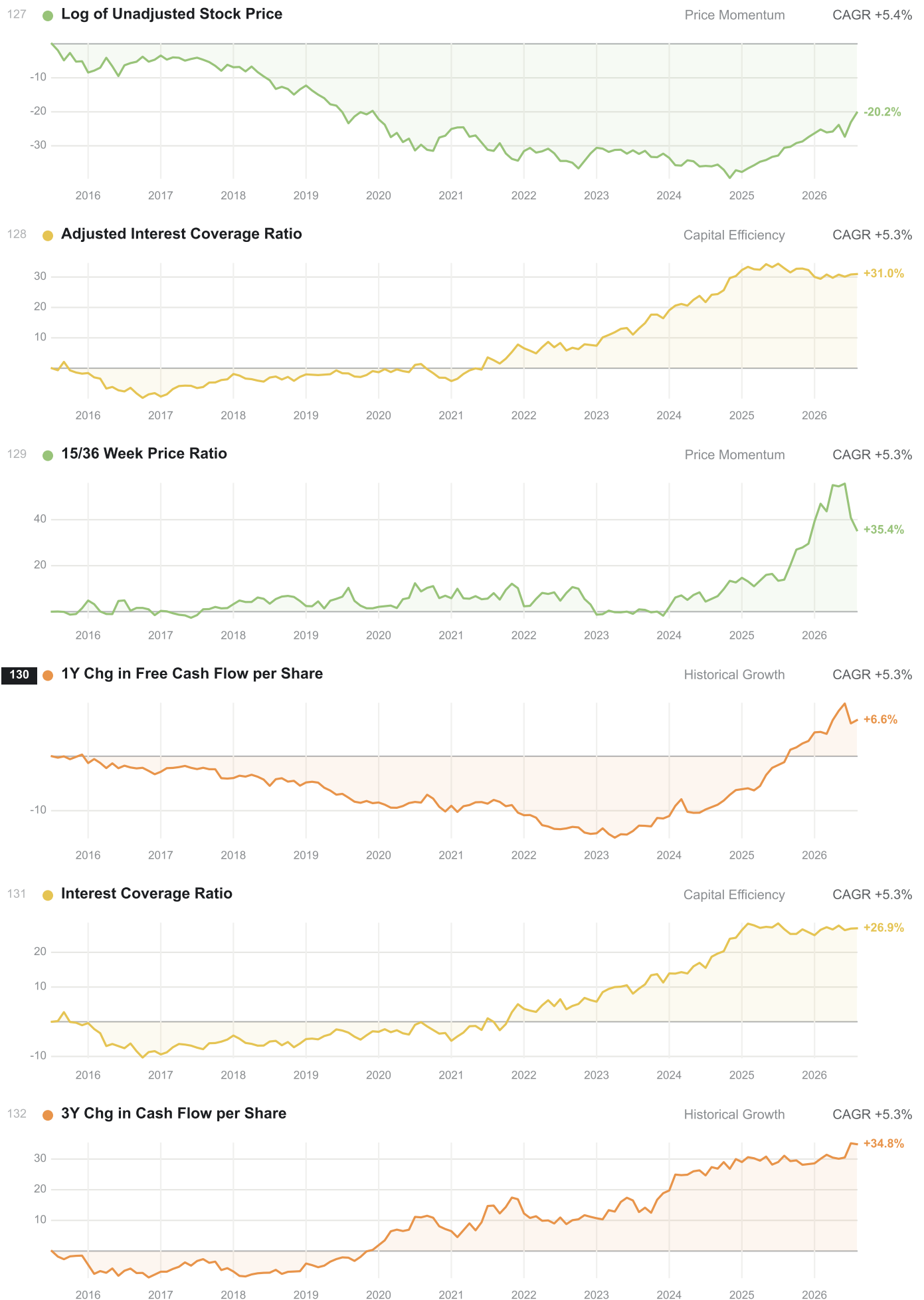
Historical Growth

CAGR +5.6%





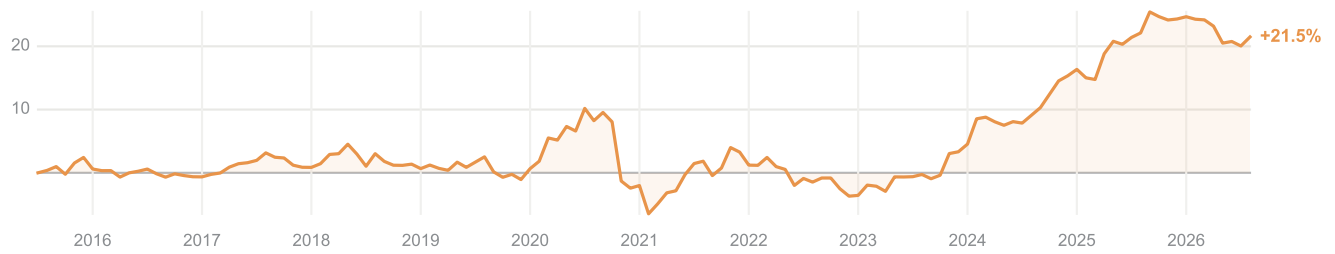




133 ● 1Y Chg in Asset Adjusted Operating Cash Flow

Historical Growth

CAGR +5.3%



134 ● Free Cash Flow Stability

Earnings Quality

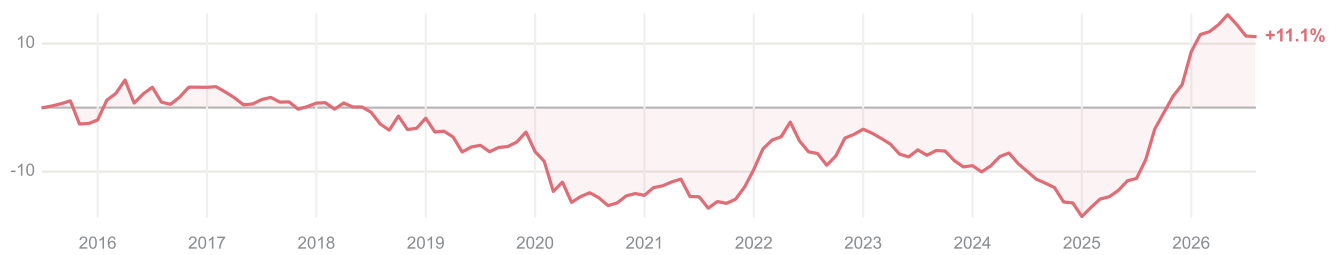
CAGR +5.3%



135 ● SG&A to Sales

Earnings Quality

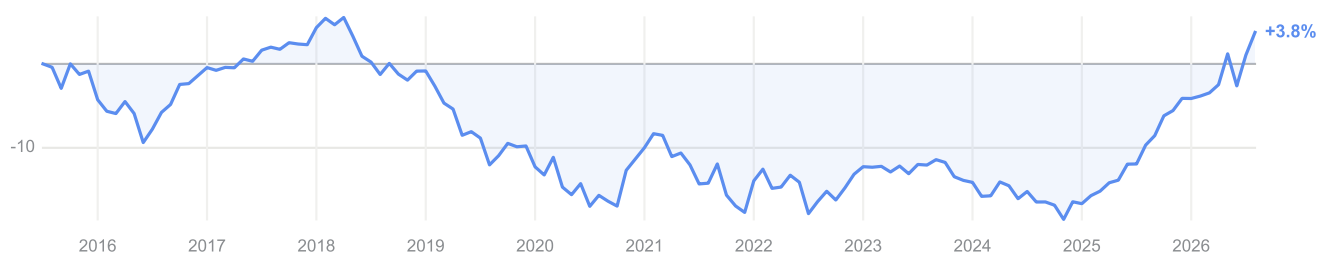
CAGR +5.3%



136 ● Cash to EV Ratio

Valuation

CAGR +5.3%



137 ● 3Y Chg in Earnings to Price

Historical Growth

CAGR +5.2%



138 ● 3Y Chg in Asset Adjusted Cash Flow

Historical Growth

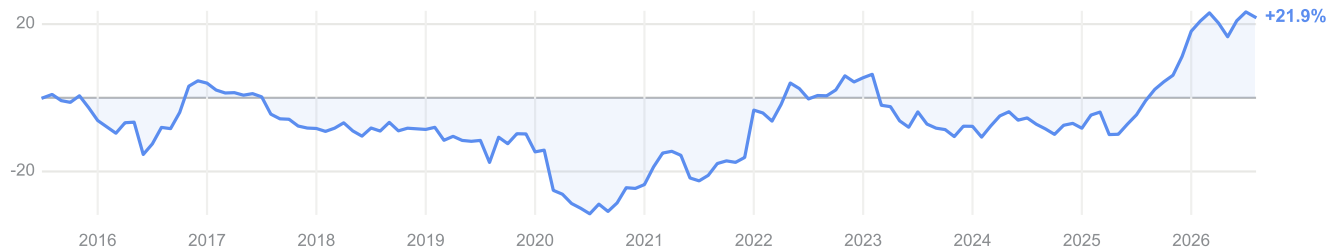
CAGR +5.2%



139 ● **Adjusted Book to Enterprise Value**

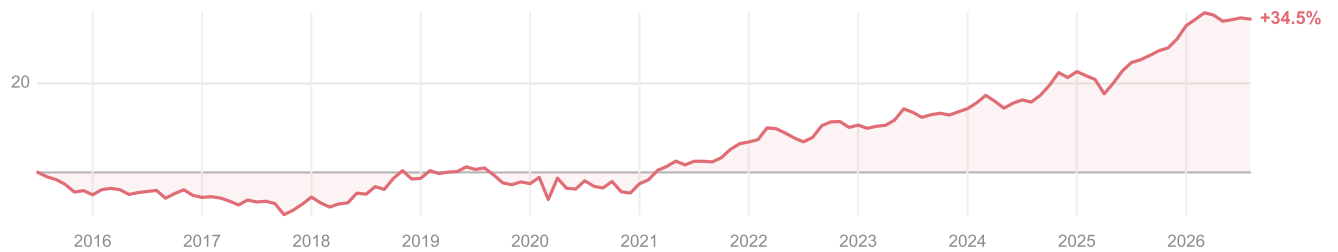
Valuation

CAGR +5.2%

140 ● **Working Capital Accruals**

Earnings Quality

CAGR +5.1%

141 ● **9 Month Price Momentum**

Price Momentum

CAGR +5.1%

142 ● **Free Cash Flow to Sales**

Earnings Quality

CAGR +5.1%

143 ● **Asset Turnover**

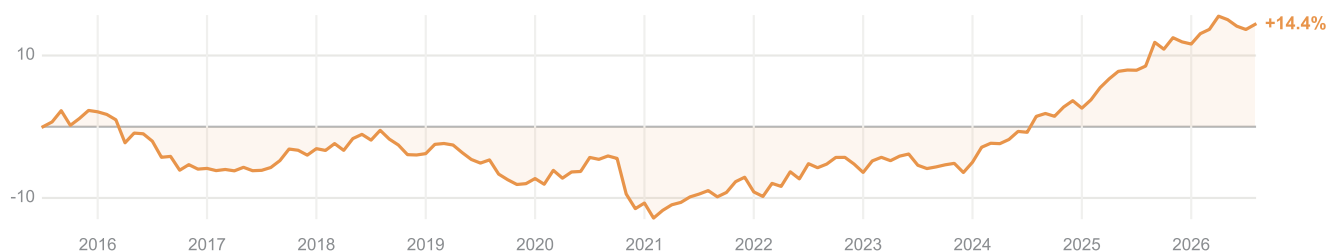
Earnings Quality

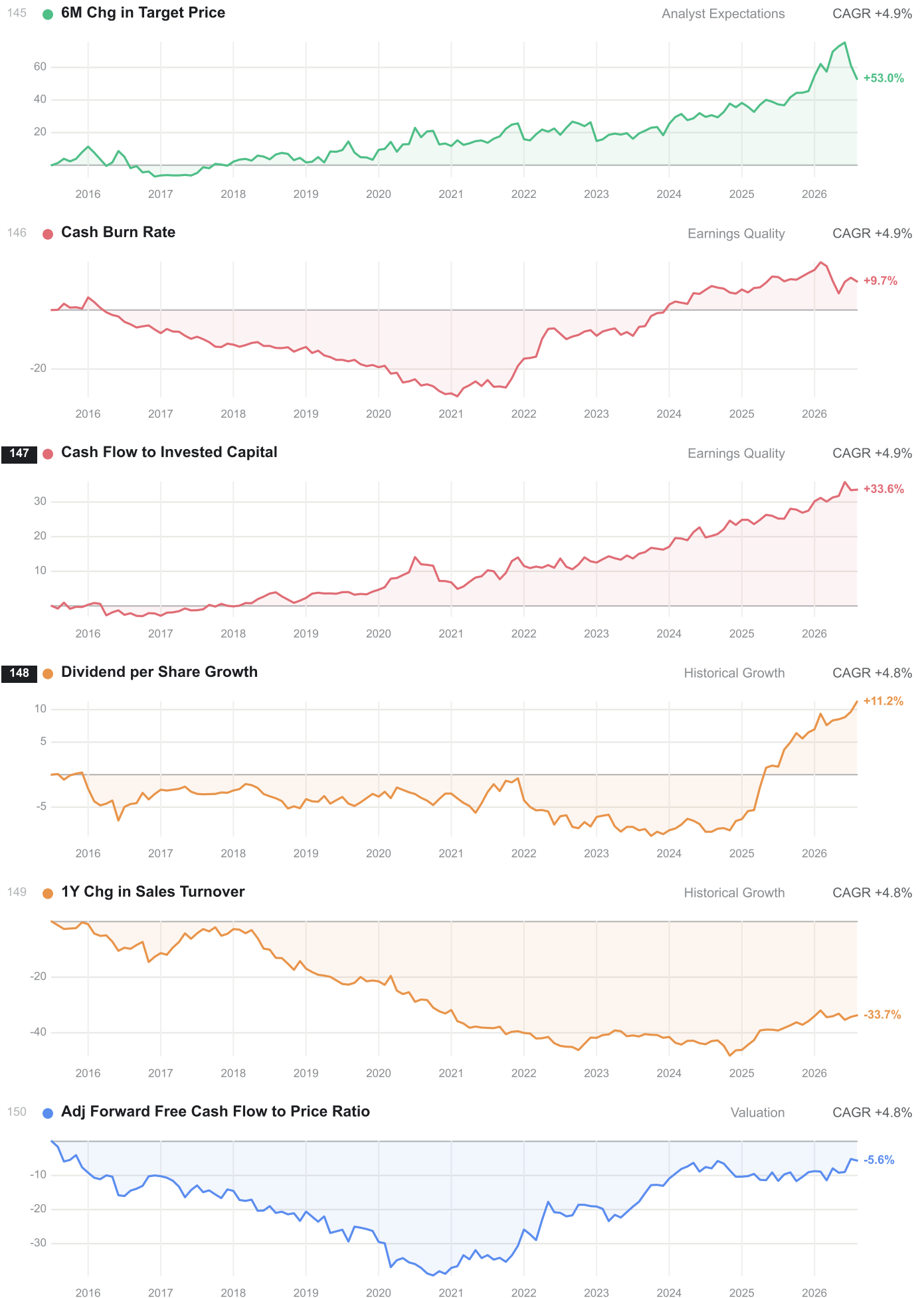
CAGR +5.1%

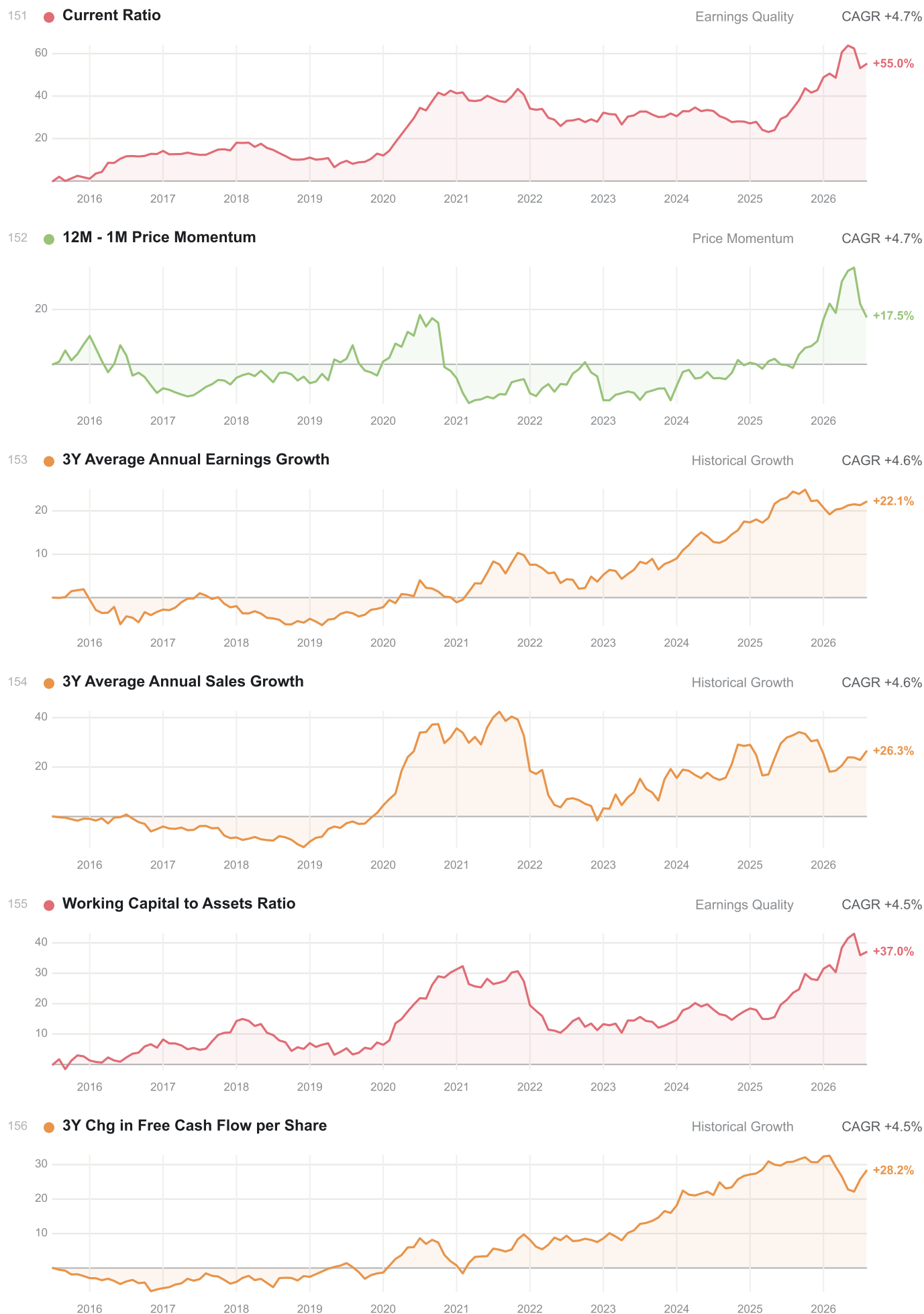
144 ● **EPS Stability**

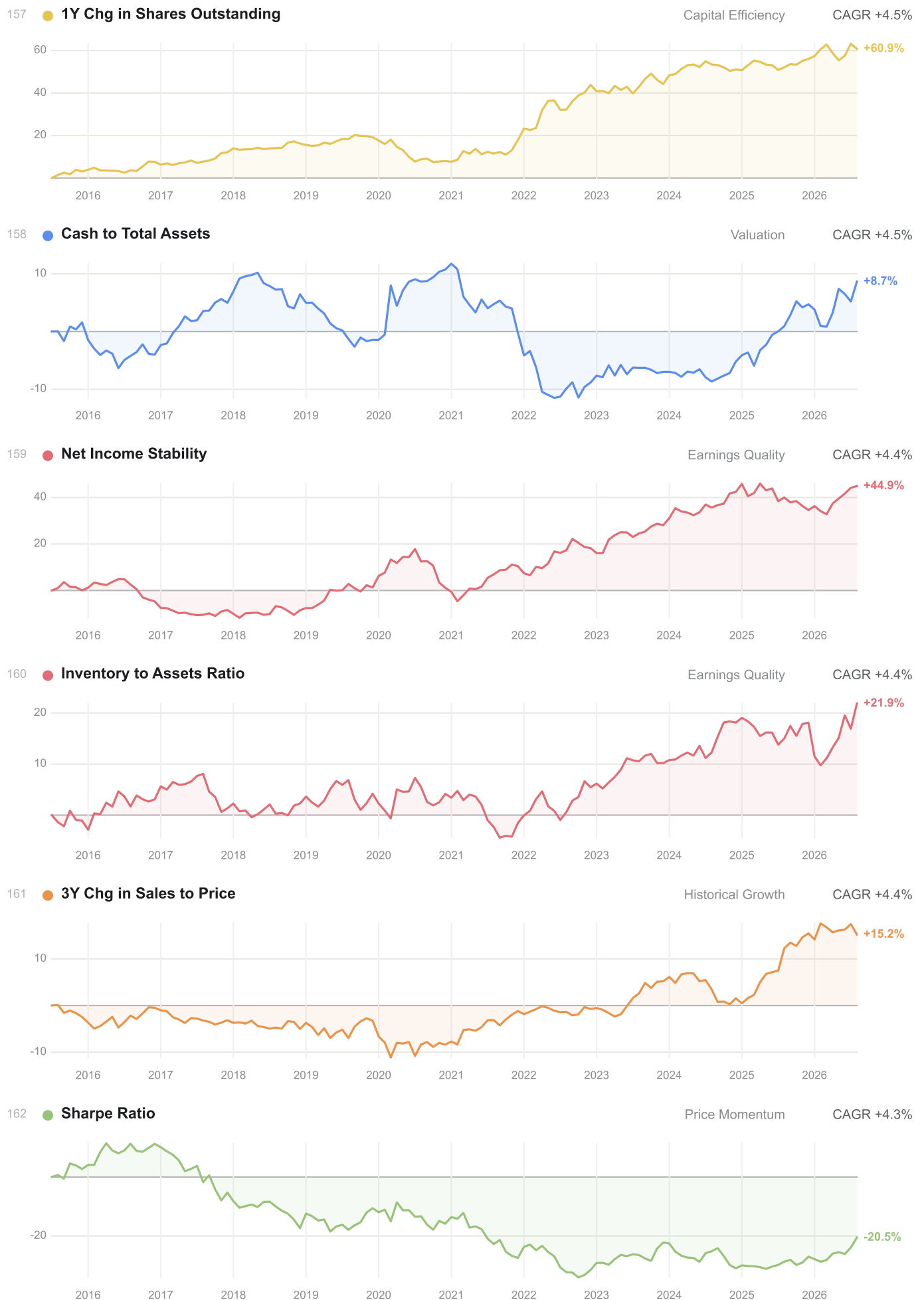
Historical Growth

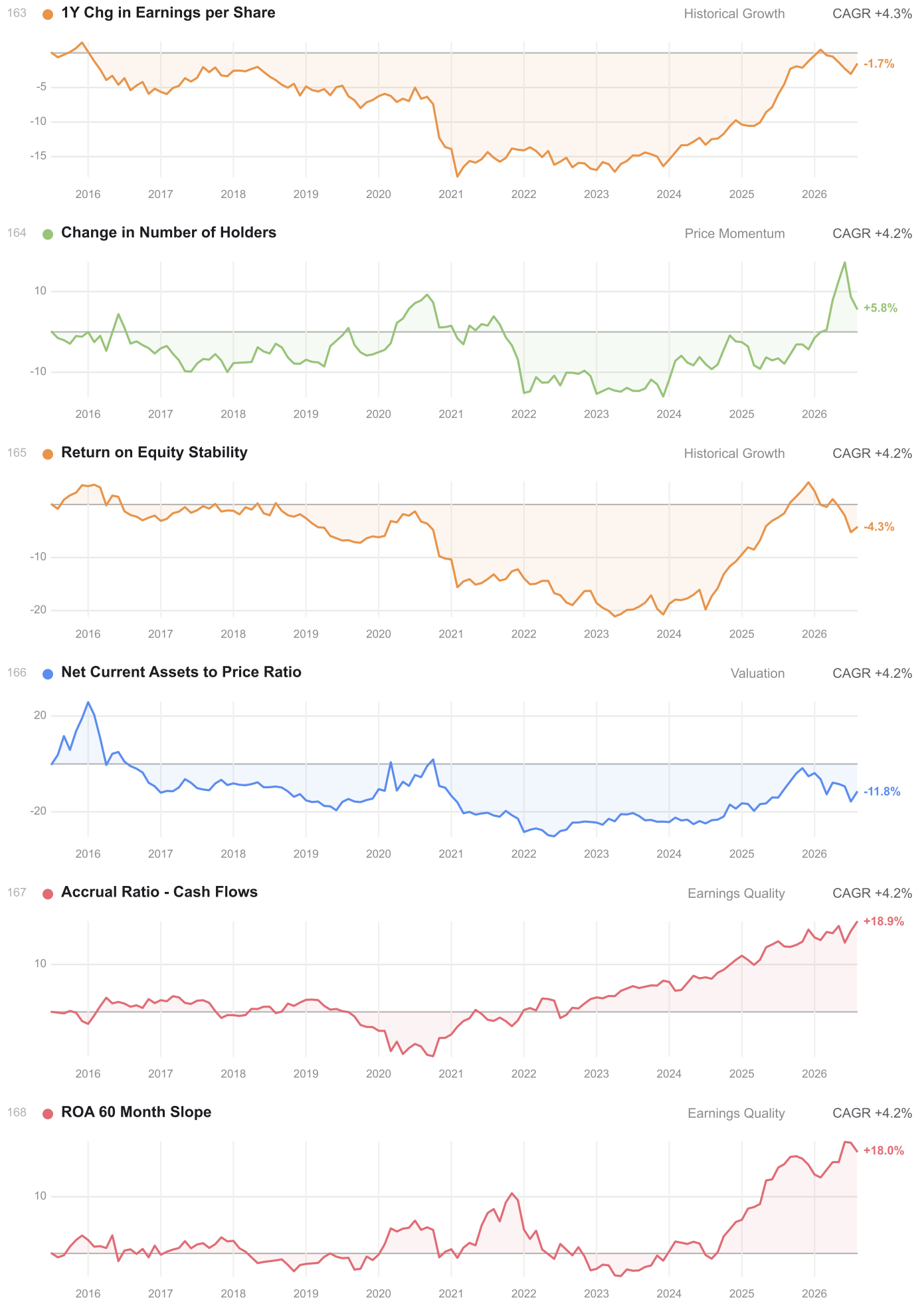
CAGR +5.0%







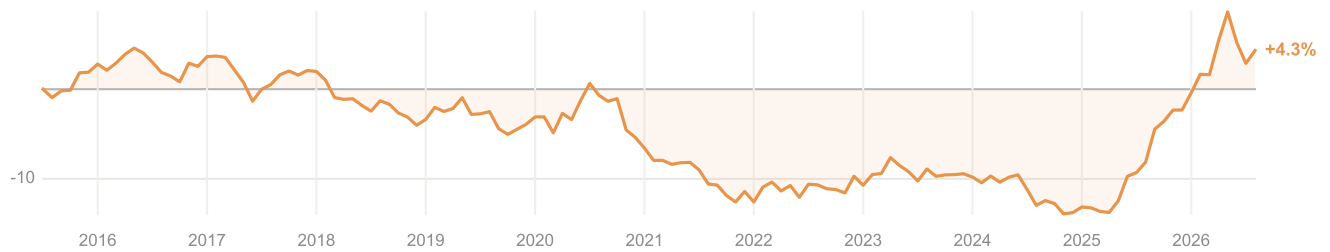




169 ● **3M Momentum in Trailing 12M Sales**

Historical Growth

CAGR +4.1%

170 ● **Quick Ratio**

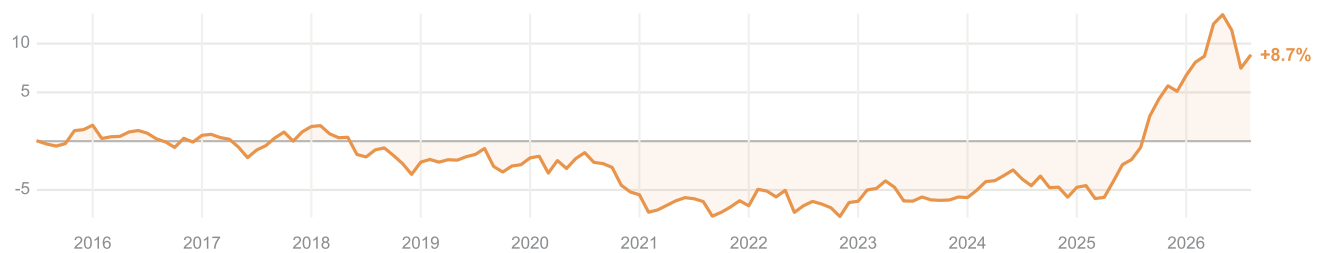
Earnings Quality

CAGR +4.1%

171 ● **6M Momentum in Trailing 12M Sales**

Historical Growth

CAGR +4.1%

172 ● **Return on Assets Stability**

Historical Growth

CAGR +4.0%

173 ● **Number of EPS FY1 Revisions**

Analyst Expectations

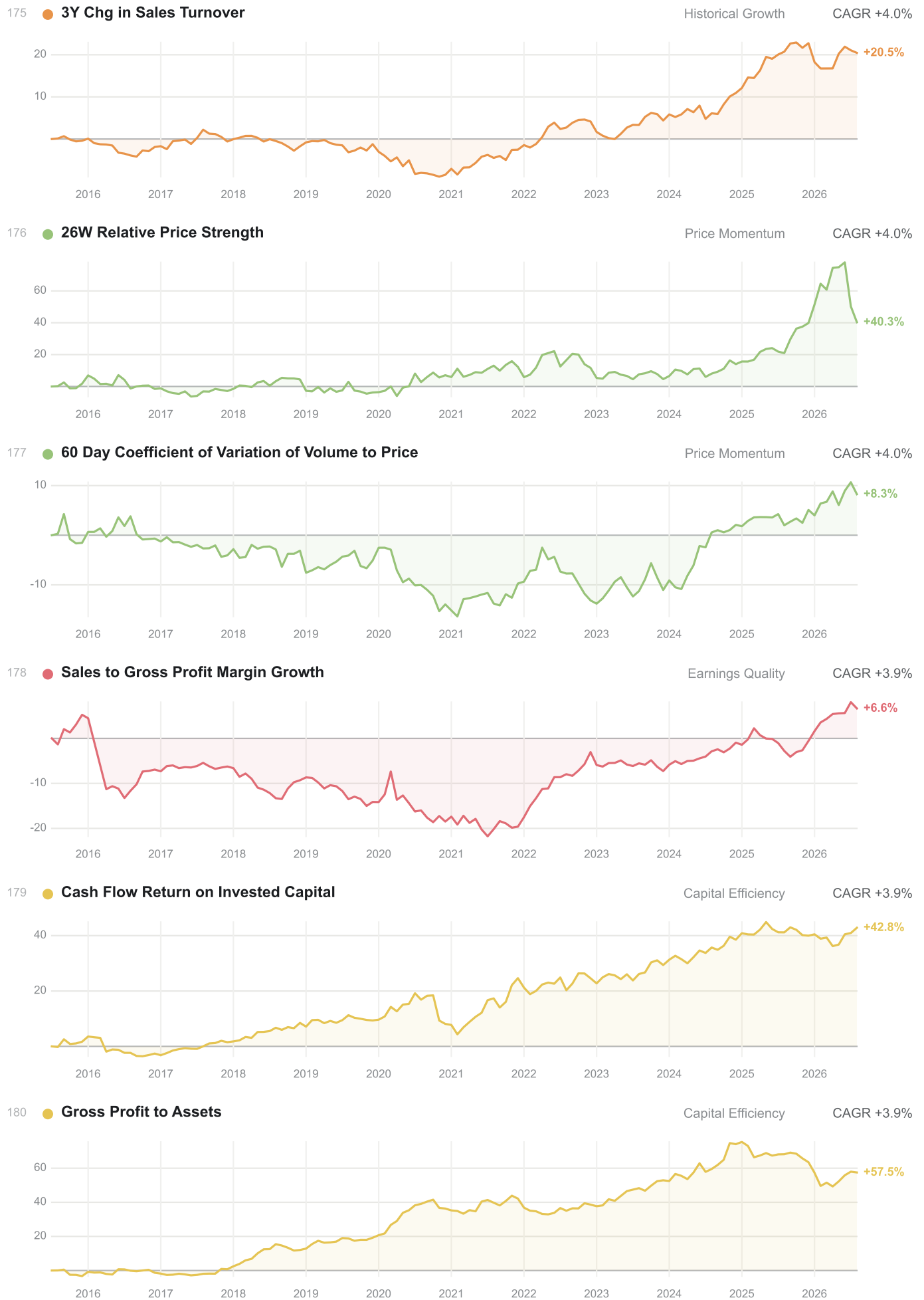
CAGR +4.0%

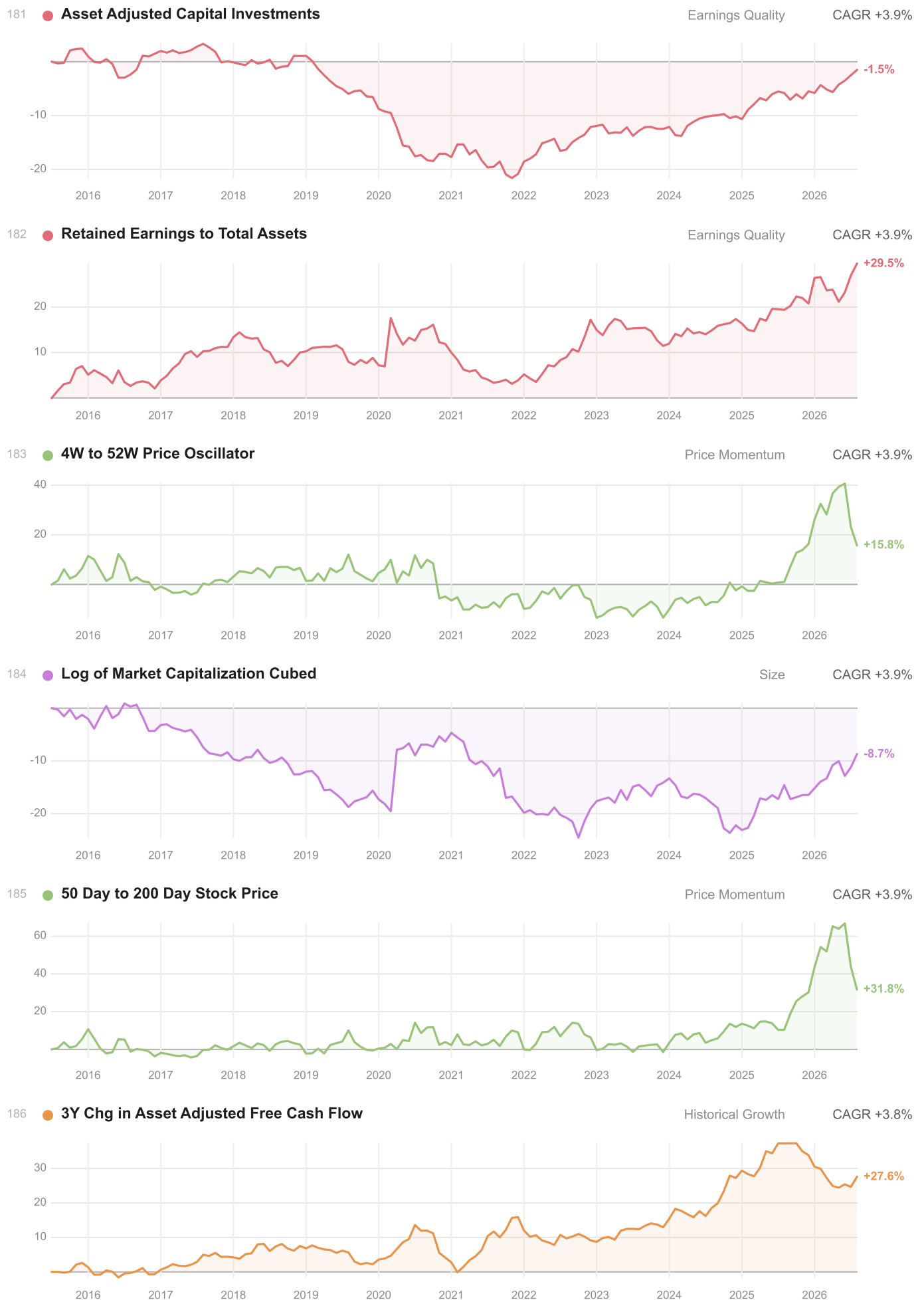
174 ● **Share Turnover**

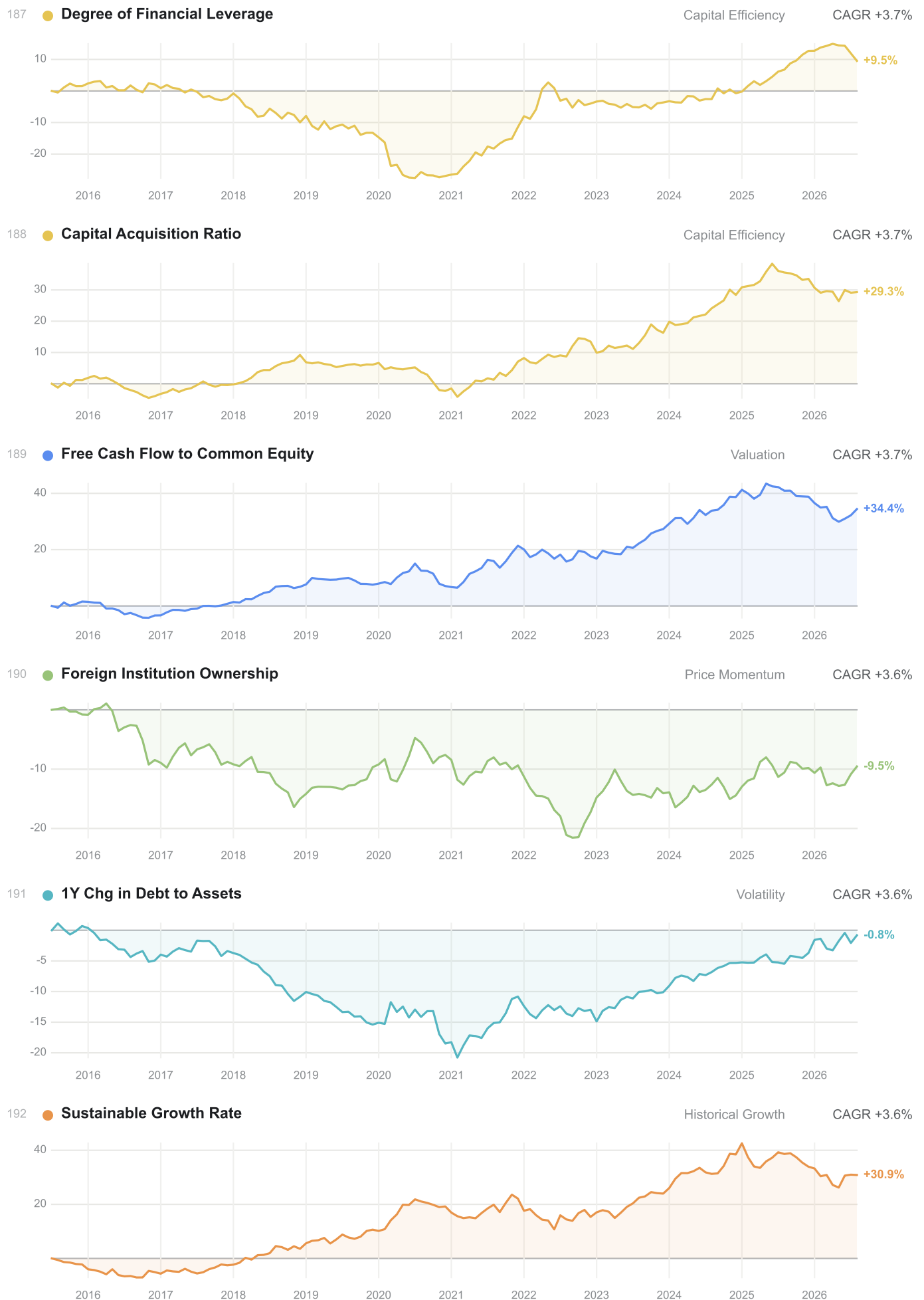
Price Momentum

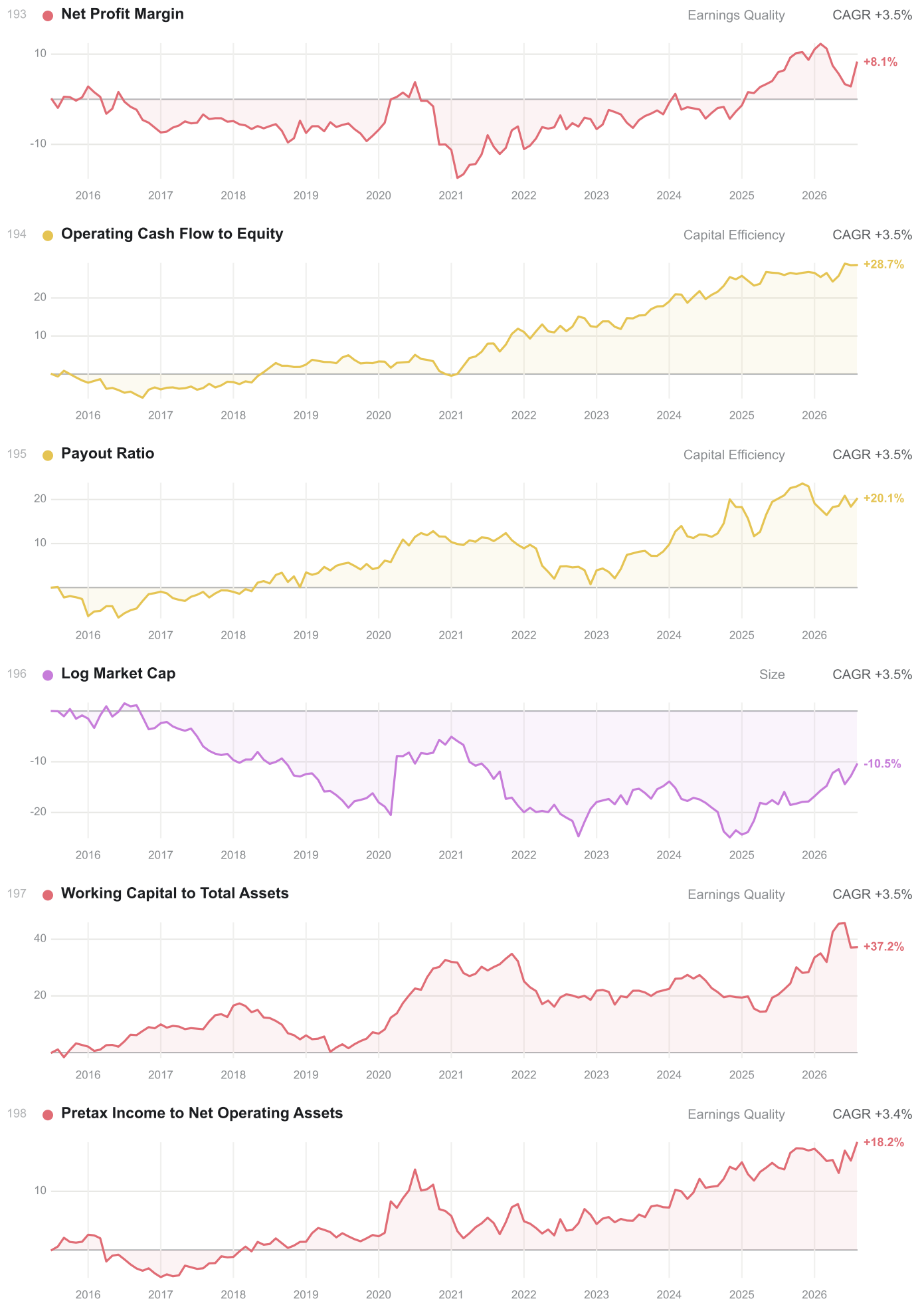
CAGR +4.0%

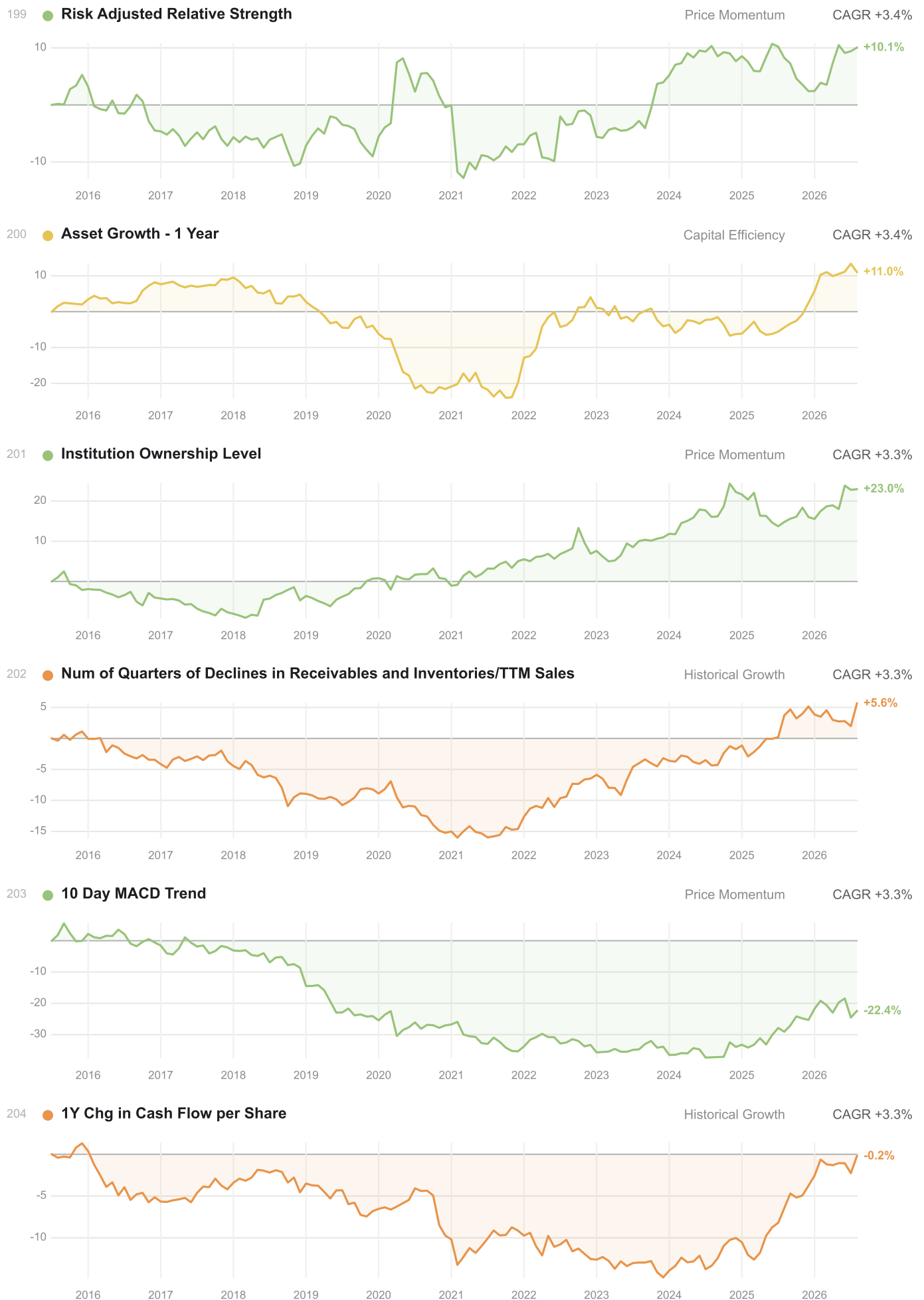








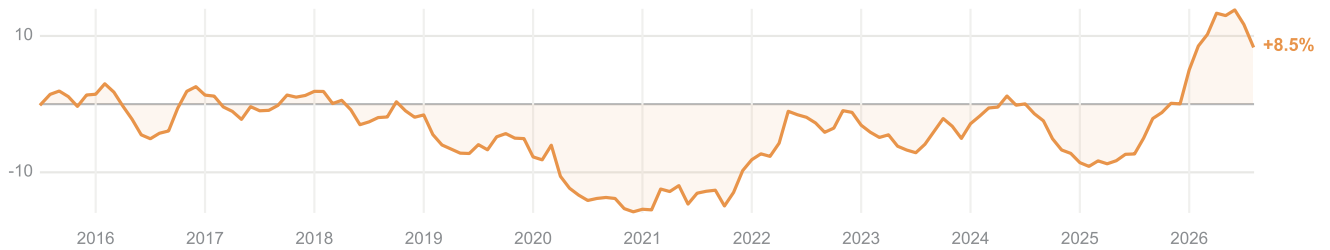




205 ● **Trailing 12-Month Overhead/Trailing 12-Month Sales**

Historical Growth

CAGR +3.3%

206 ● **3Y Chg in Earnings per Share**

Historical Growth

CAGR +3.2%

207 ● **1Y Chg in Sales to Earnings**

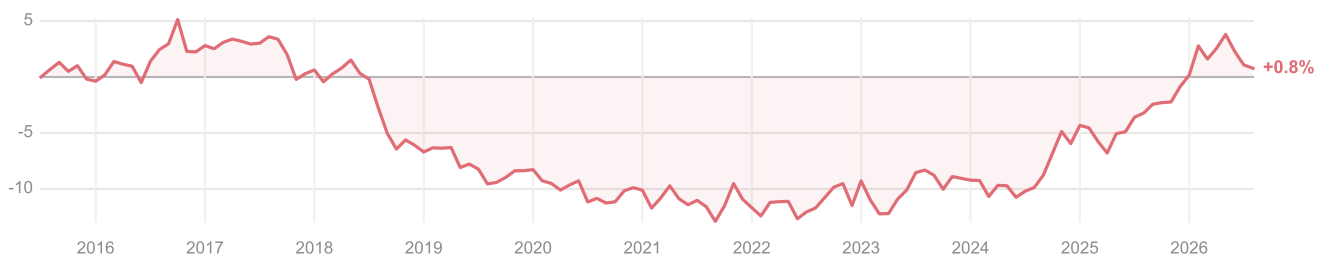
Earnings Quality

CAGR +3.2%

208 ● **Unexpected Receivables Change**

Earnings Quality

CAGR +3.2%

209 ● **1Y Chg in EPS to Operating Cash Flow**

Earnings Quality

CAGR +3.2%

210 ● **6M Chg in 18M CAPM Alpha**

Price Momentum

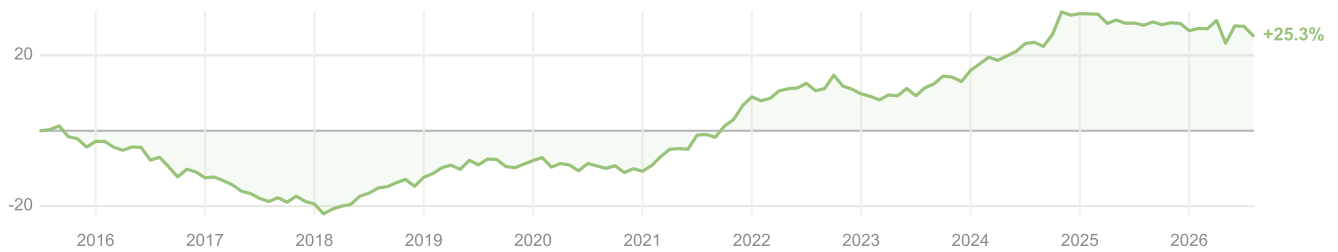
CAGR +3.2%



211 ● **Institution Ownership Concentration**

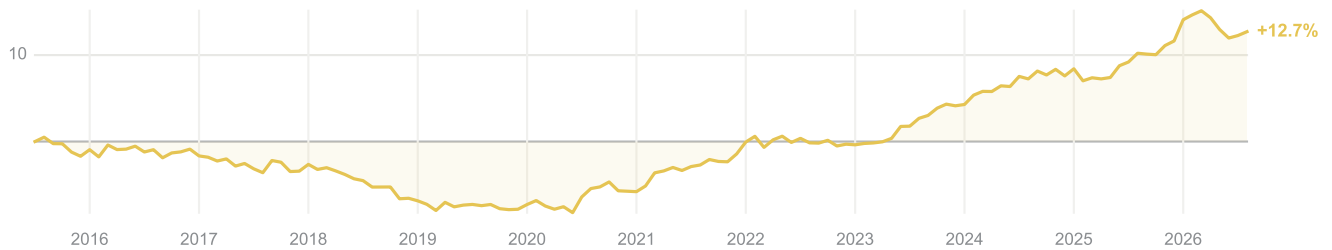
Price Momentum

CAGR +3.1%

212 ● **Working Capital Turnover Ratio**

Capital Efficiency

CAGR +3.1%

213 ● **Target Price Gap to 6M EMA**

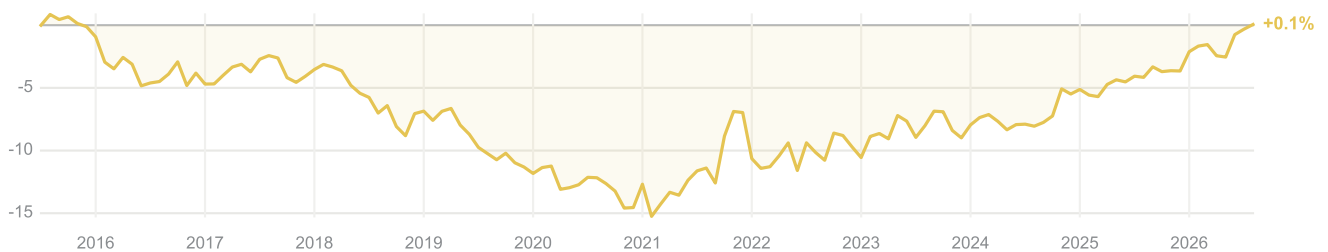
Analyst Expectations

CAGR +3.1%

214 ● **Year over Year Change of Total Debt**

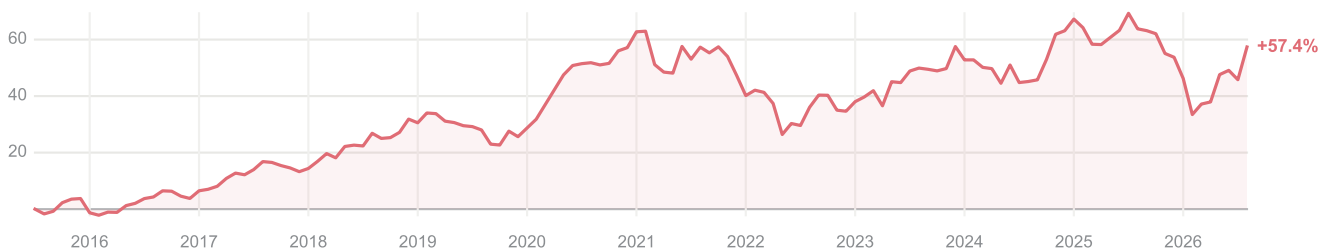
Capital Efficiency

CAGR +3.0%

215 ● **Research & Development Intensity**

Earnings Quality

CAGR +3.0%

216 ● **Cash Conversion Cycle**

Earnings Quality

CAGR +2.9%



217 ● **6M Chg in 36M CAPM Alpha**

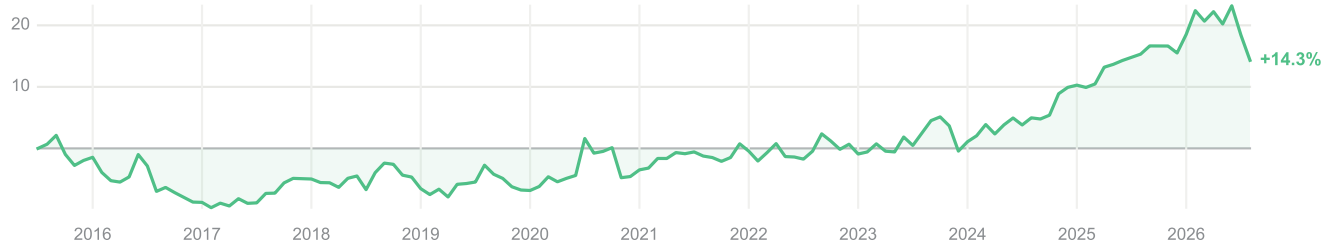
Price Momentum

CAGR +2.9%

218 ● **Adj Number of EPS FY2 Revisions**

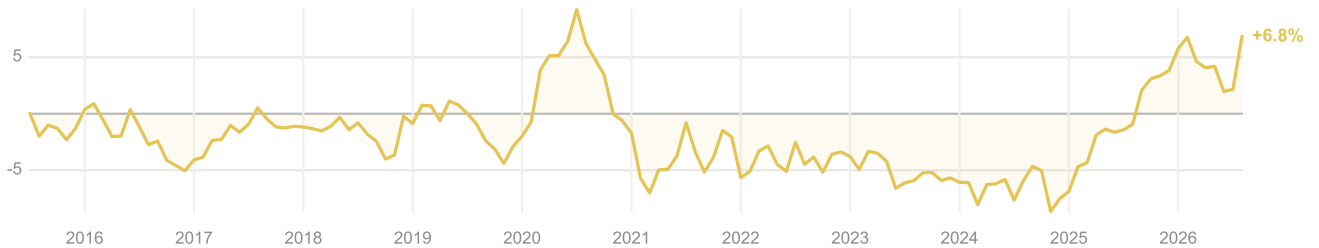
Analyst Expectations

CAGR +2.8%

219 ● **Pretax Profit Margin**

Capital Efficiency

CAGR +2.8%

220 ● **Accrual Ratio - Balance Sheet**

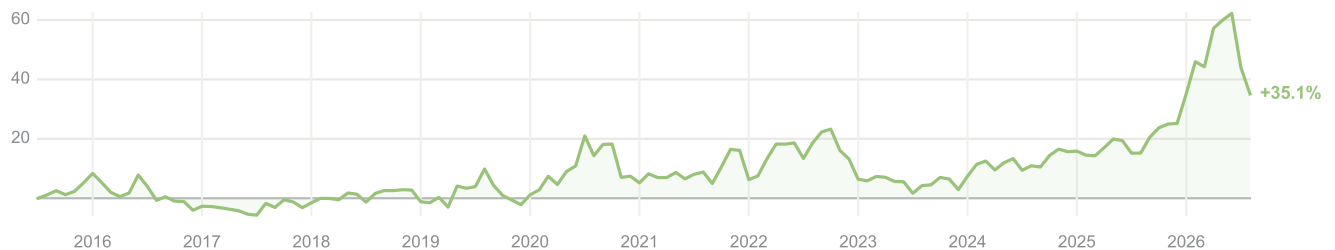
Earnings Quality

CAGR +2.7%

221 ● **180D Price TStat**

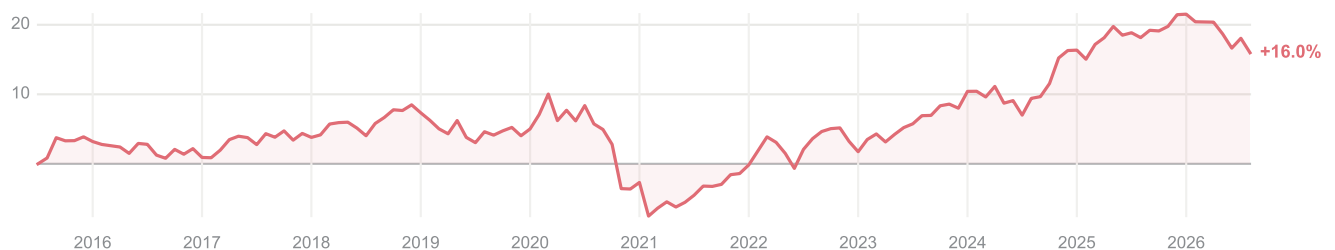
Price Momentum

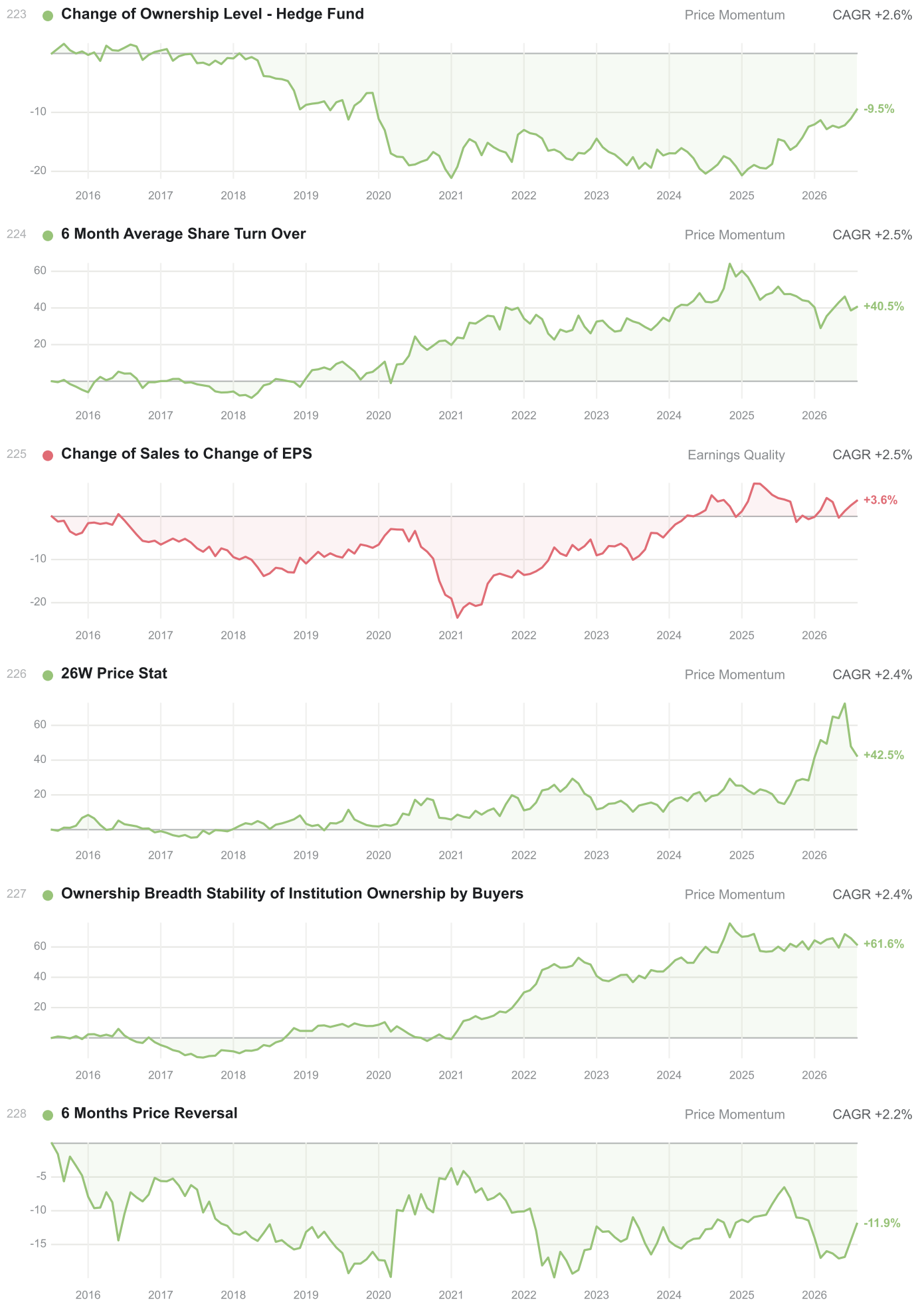
CAGR +2.6%

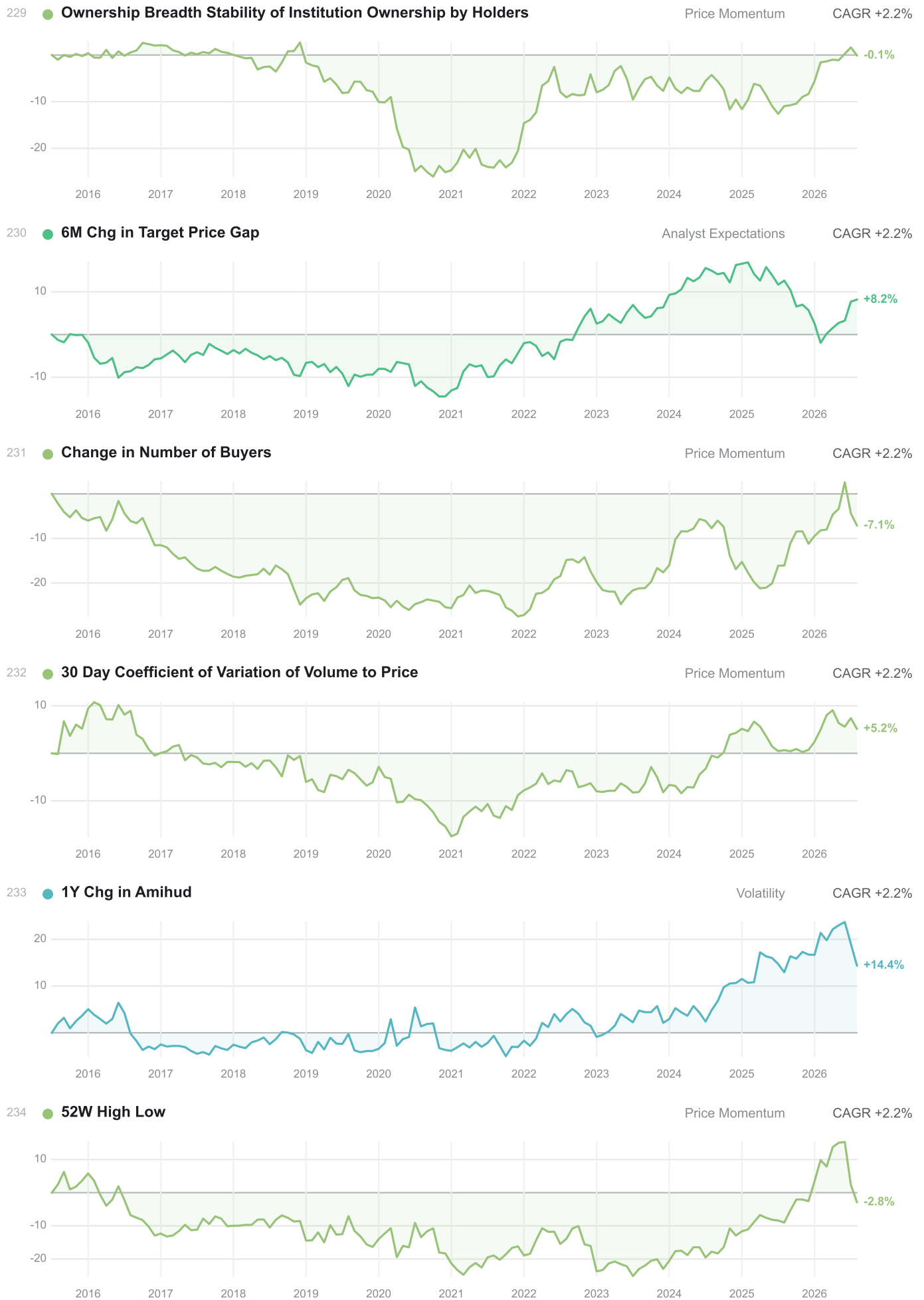
222 ● **Operating Cash Flow Stability**

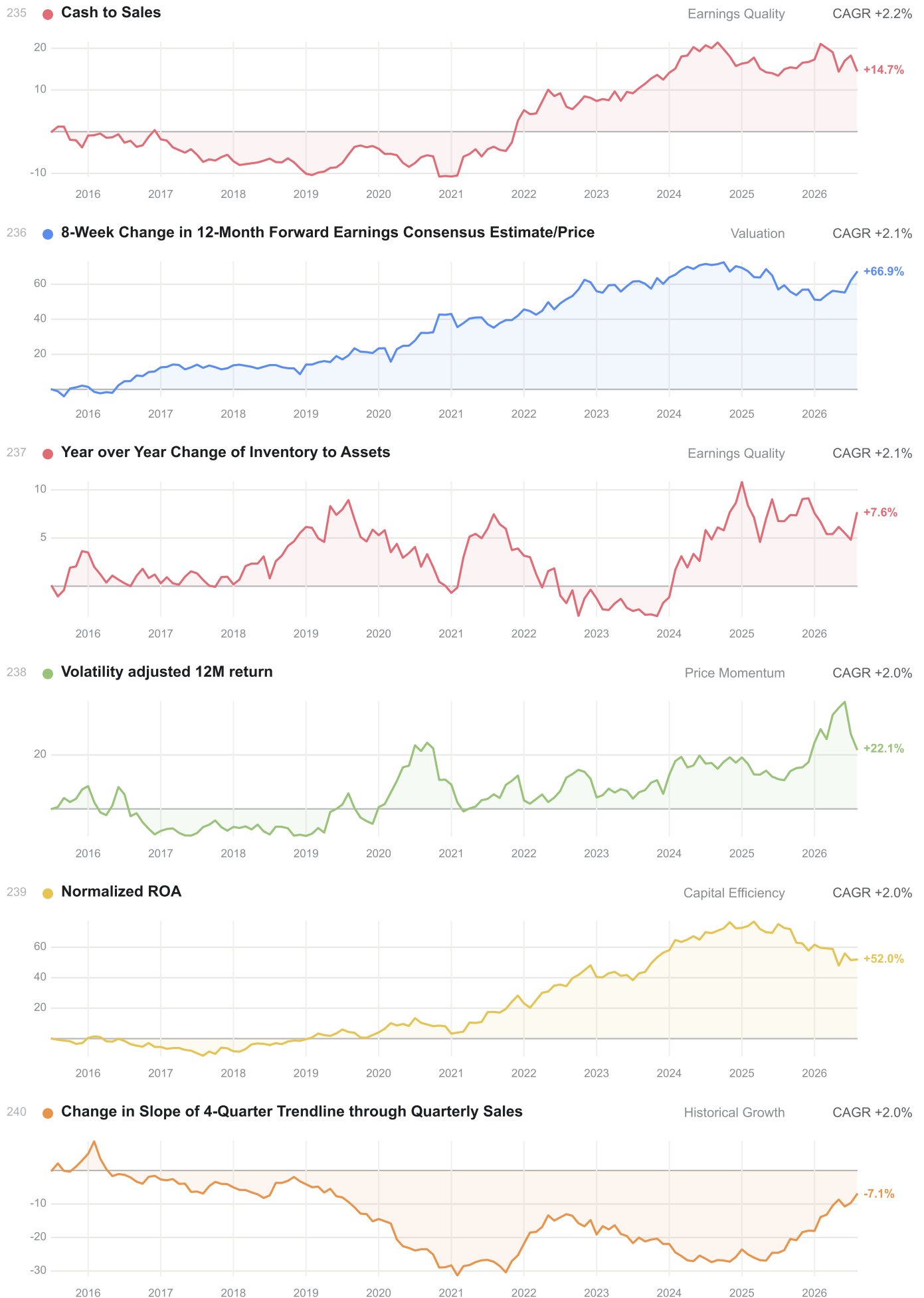
Earnings Quality

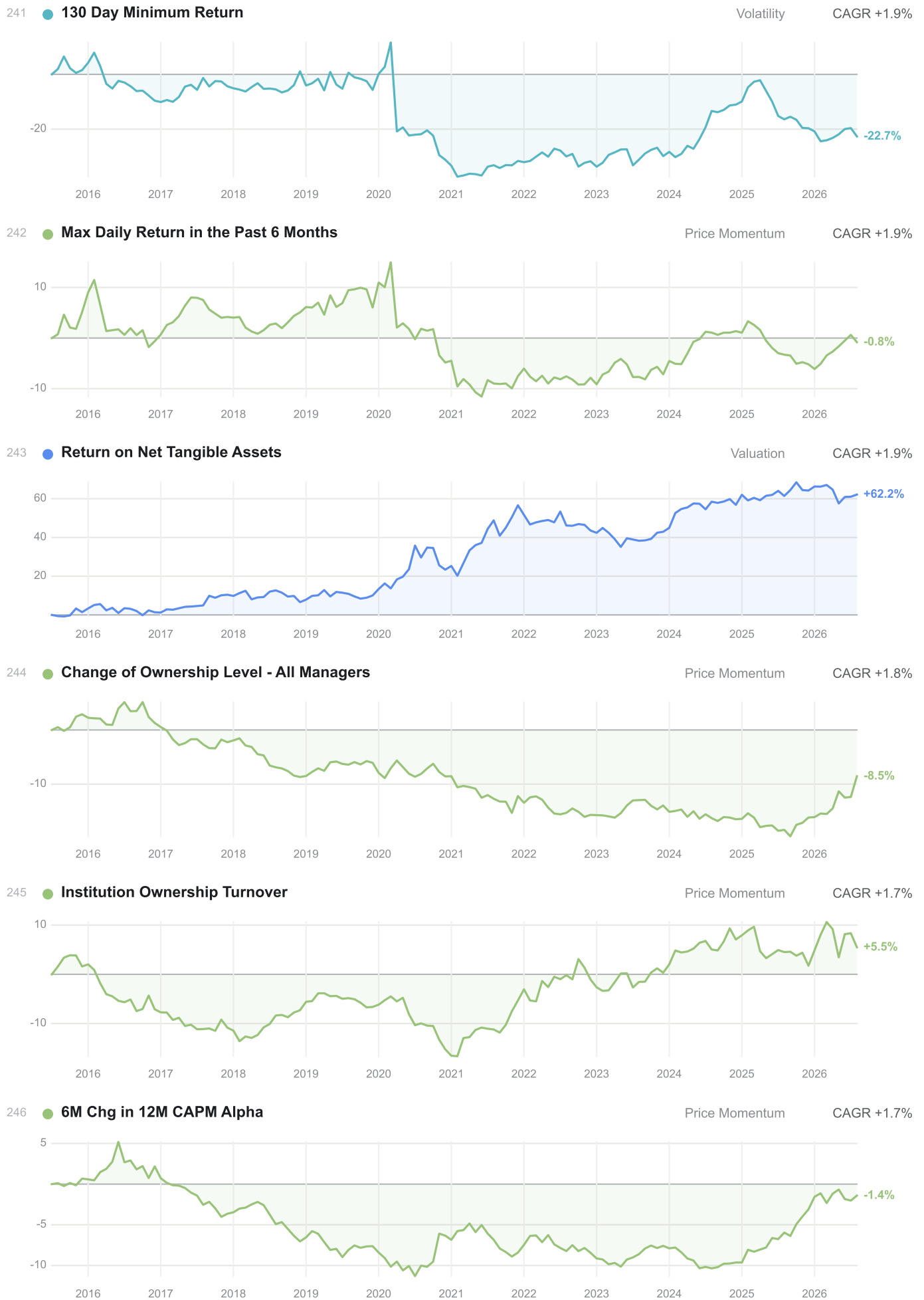
CAGR +2.6%

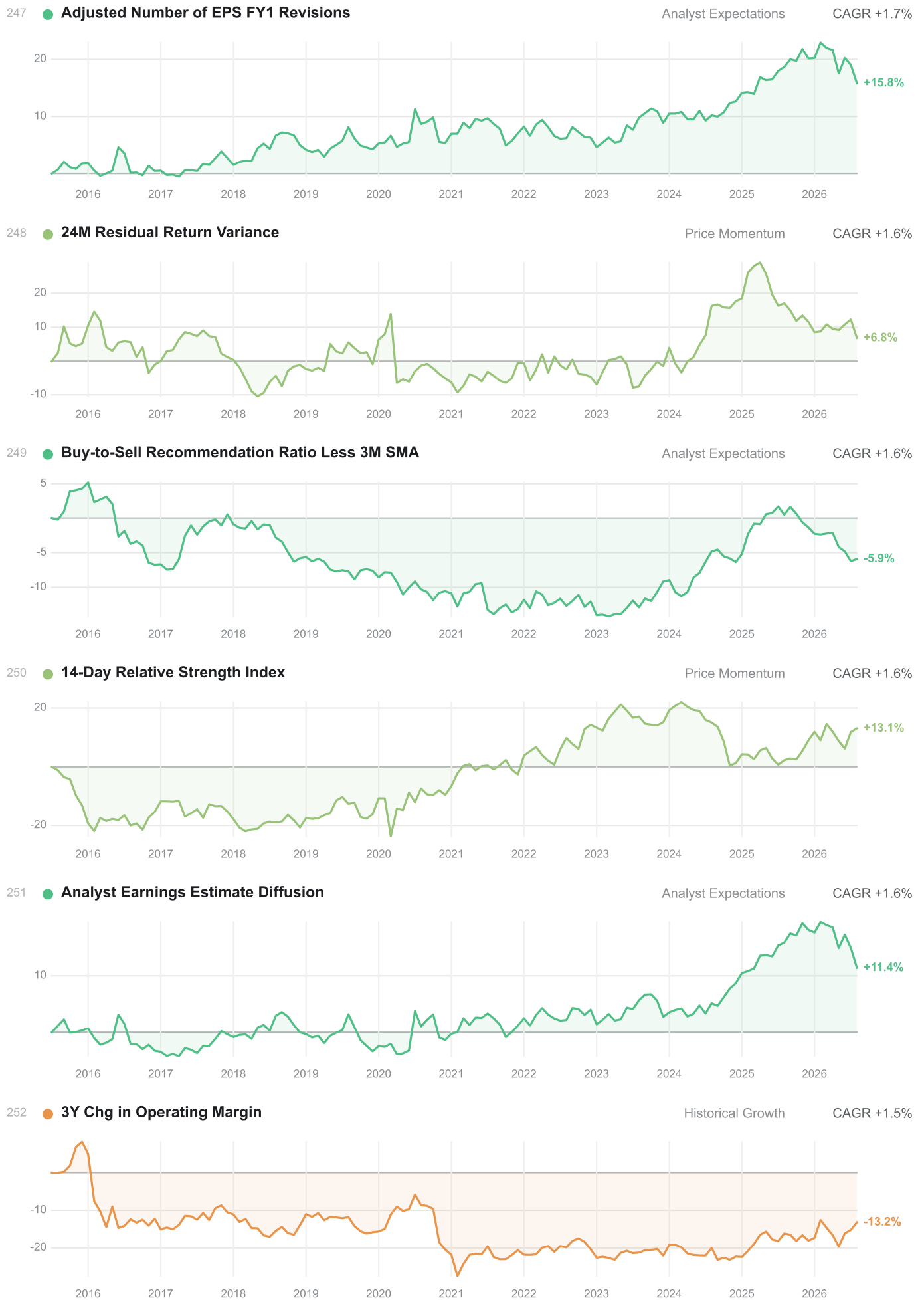








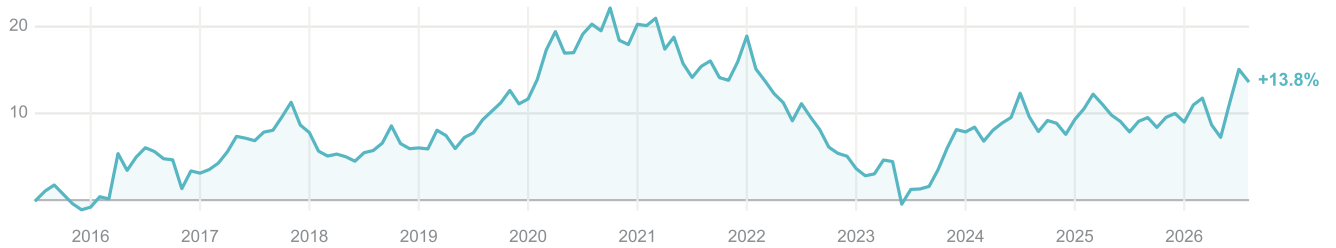




253 ● **Adj 50 Day Volume Signal**

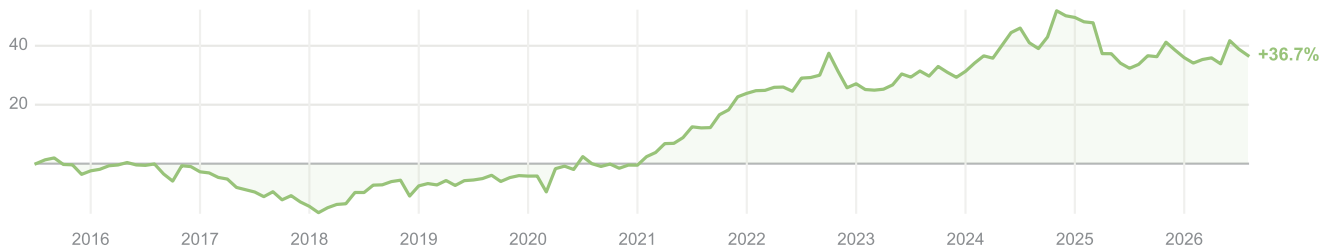
Volatility

CAGR +1.5%

254 ● **Investment Duration**

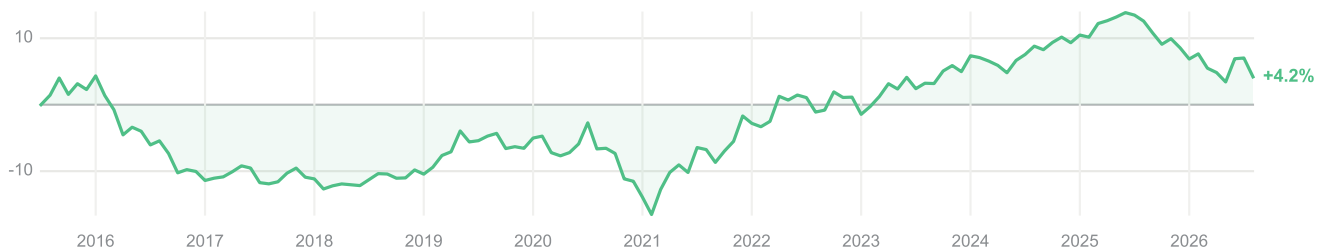
Price Momentum

CAGR +1.5%

255 ● **EBIT Estimate Dispersion**

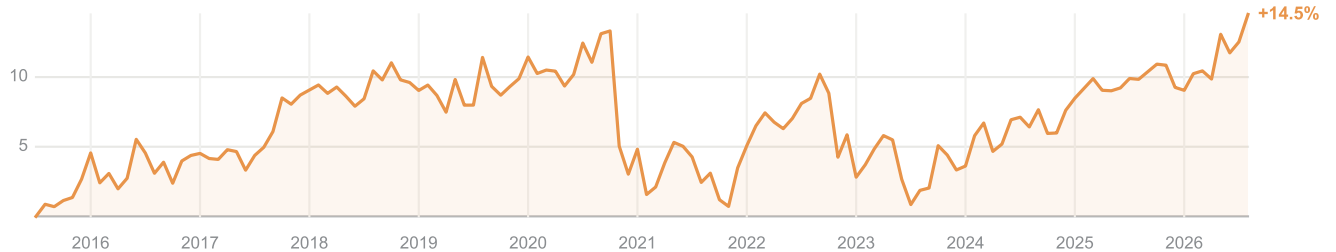
Analyst Expectations

CAGR +1.4%

256 ● **Sales Acceleration**

Historical Growth

CAGR +0.9%

257 ● **Revenue Estimate Dispersion**

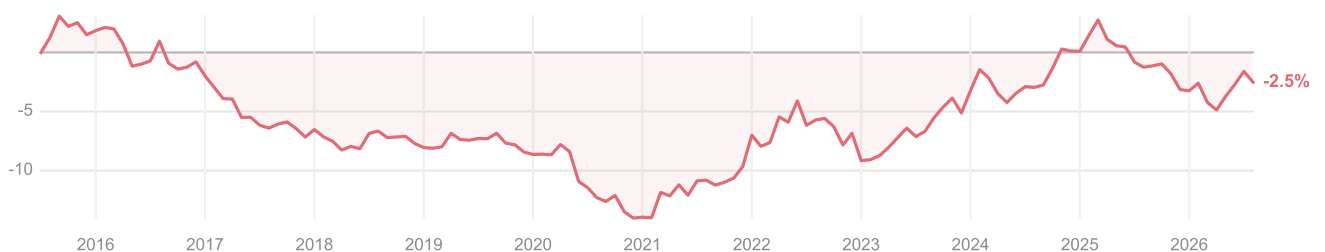
Analyst Expectations

CAGR +0.8%

258 ● **Change in TTM Depr. to Capex**

Earnings Quality

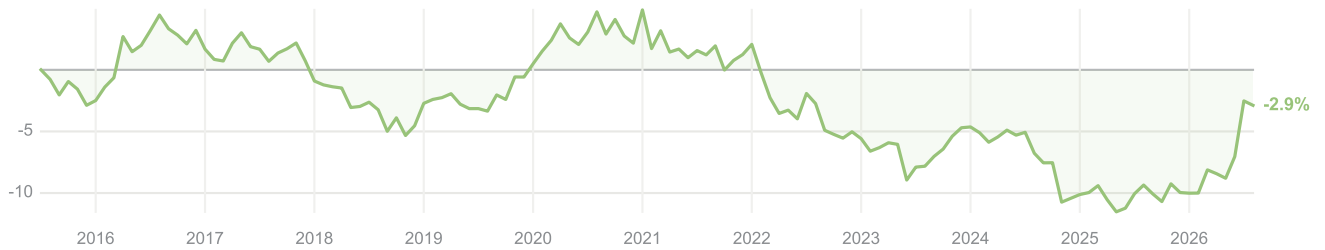
CAGR +0.8%



259 ● 50 Day Volume Signal

Price Momentum

CAGR +0.7%



260 ● Buy-to-Sell Recommendation Ratio Less 3M EMA

Analyst Expectations

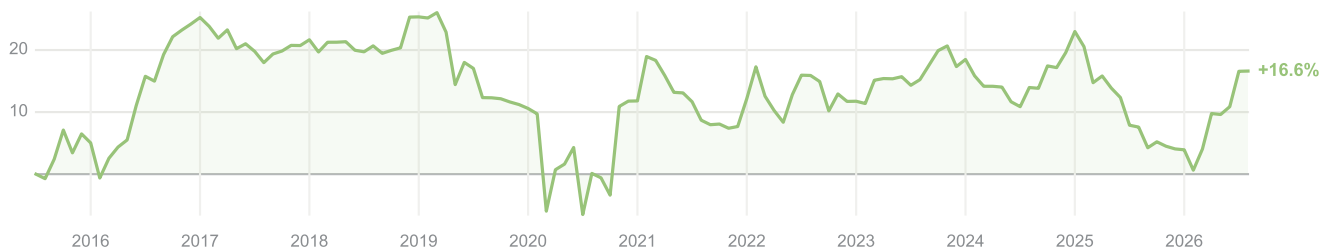
CAGR +0.6%



261 ● 5 Day Price Reversal

Price Momentum

CAGR +0.6%



262 ● Closing Price to 52 Week High

Price Momentum

CAGR +0.6%



263 ● ROE 20 Qtr Standard Deviation

Earnings Quality

CAGR +0.5%

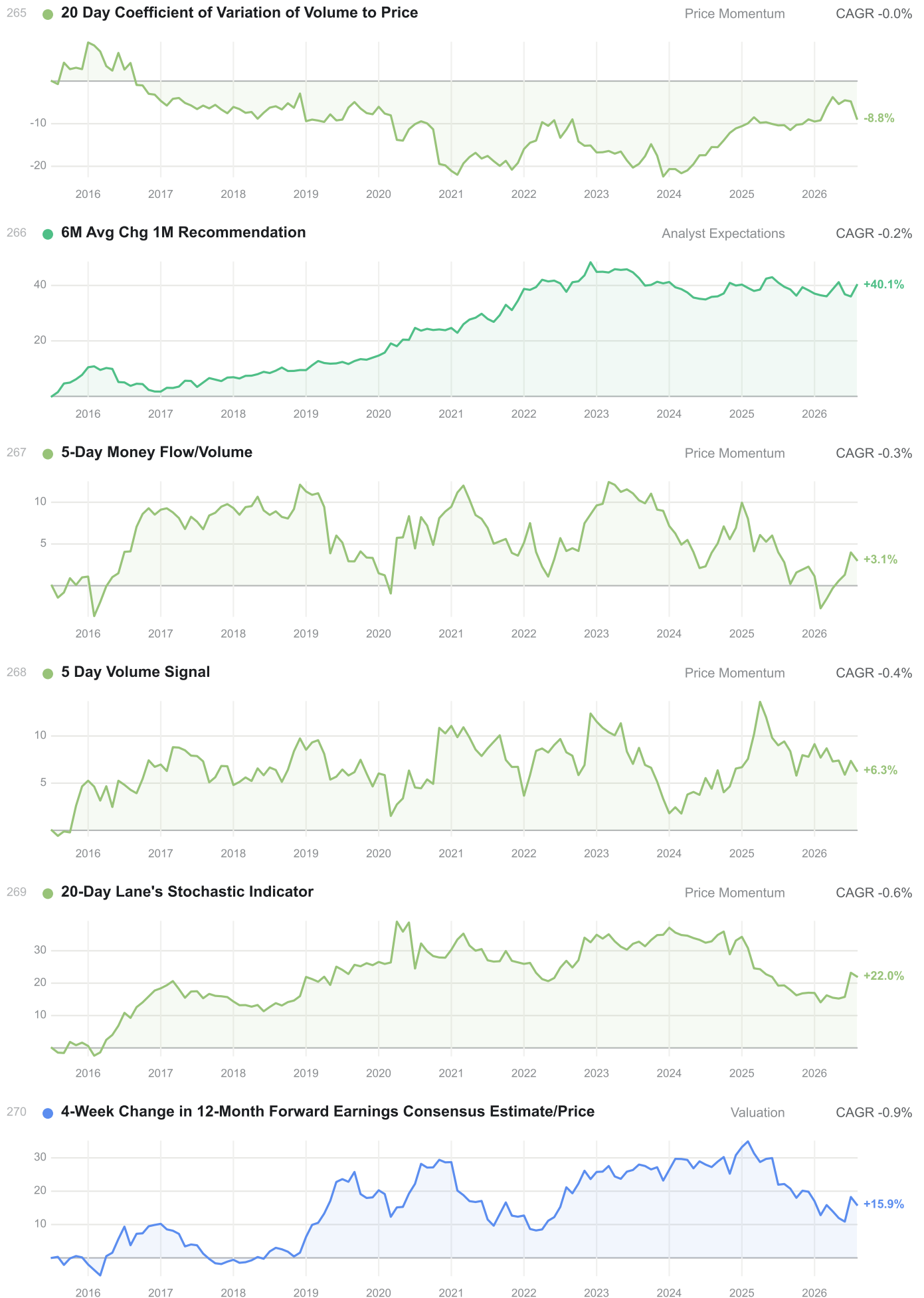


264 ● 1Y Chg in Share Turnover

Price Momentum

CAGR +0.3%





271 ● **Normalized ROE**

Capital Efficiency

CAGR -1.0%

272 ● **1M Price High - 1M Price Low**

Price Momentum

CAGR -1.3%

